Machine-checked Weingarten calculus: the unitary, orthogonal, and symplectic packages with the U(N) Haar-integration bridge

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Chapter 1

Introduction and scope

This blueprint specifies a Lean 4 / Mathlib formalization of the *stable-range Weingarten package*: the Jucys factorization of the Gram element of the symmetric-group algebra; its invertibility for real N with $n-1 < N$ via an ℓ^1 Neumann series; the rising and falling sum rules; the strict parity sign theorem; the exact closed form for the cancellation ratio; and a certified below-threshold cell at $(N, n) = (2, 3)$. A below-threshold package (Chapter 4) states both sum rules *conditionally on the Penrose equations* — hence at every N, with no irreducible-character theory — including the exact vanishing of the signed sum for integer $1 \leq N < n$, the forced breakdown of the sign pattern, and the singularity of the Gram element, with the cell of Chapter 9 as the certified existence instance. Chapter 13 states the $\mathrm{U}(N)$ Haar-integration bridge at the formalized degrees.

Every theorem-class statement is known mathematics or an elementary corollary of known mathematics (the five prose remarks carry their own status flags, one of them an explicitly marked conjecture); the paper proofs are recorded in this blueprint, and the `scripts/` directory — driven by `scripts/verify_all.py` — cross-checks every finite computational theorem by algorithmically independent recomputation in exact rational arithmetic (within this repository; not third-party verification), exhaustively over all cases up to a size bound, with the analytic statements (the ℓ^1 Neumann invertibility and the `tsum` expansions) exercised through their finite numeric consequences. The deliberate scope boundary — what is and is not formalized, and why — is recorded in Chapter 14. The sources of record are public at https://github.com/SciSolv/weingarten-lean (pinned Lean/Mathlib toolchain, the verification suites, continuous-integration gates, and this blueprint's LaTeX source; the release version is carried by `CITATION.cff`).

Status (current): the unitary spine (parity sign theorem, closed-form cancellation ratio, certified $(2, 3)$ cell), the certified Penrose pair, and the *entire* orthogonal Gram package are formalized *sorry*-free and axiom-clean (`#print axioms` = `propext`, `Classical.choice`, `Quot.sound`; no `sorryAx`, no `native_decide`). The symplectic Gram package is likewise *fully* proved and axiom-clean (conjugation, matrix factorization, both ε-twisted absorptions, the $J^2 = -1$ trace, the transfer-matrix per-loop trace, and the complete crossing-sign factorization `thm:crossing_sign` — the merge-induction, both case residuals, and the final crossing-parity lemma `qStar_sign`), as is the $\mathrm{U}(N)$ Haar-integration bridge of Chapter 13. **Every theorem-class node below carries `leanok` (the five prose remarks carry no proof obligation); the whole development is *sorry*-free and axiom-clean, independently cross-checked by the `scripts/*.py` exact-arithmetic suites (entry point `scripts/verify_all.py`; all PASS).** The Lean files are the source of record; `scripts/VERIFICATION.md` carries the live per-theorem ledger, and `scripts/*.py` (entry point `scripts/verify_all.py`) provide independent exact-arithmetic cross-checks of every finite computational theorem, exhaustively over all cases up to a size bound.

Design record: Mathlib reconnaissance of 11 June 2026

Three findings shape the design and are recorded here so future contributors need not re-litigate them.

Fact 1 (decomposition at 0**).** Mathlib provides `Equiv.Perm.decomposeFin`, an equivalence $\mathrm{Perm}(\mathrm{Fin}(n{+}1)) \simeq \mathrm{Fin}(n{+}1) \times \mathrm{Perm}(\mathrm{Fin}\, n)$ splitting off the image of 0 (not of the last element), with simp lemmas `decomposeFin_symm_apply_zero` and `decomposeFin_symm_apply_succ`. This blueprint therefore adopts the *dual* Jucys–Murphy convention $K_i := \sum_{j>i}(i\,j)$ (Definition 3), so the keystone induction (Theorem 9) peels the factor $(N + K_0)$ exactly along `decomposeFin`. See Remark 4.

Fact 2 (unattributed). No formalization of Jucys–Murphy elements or of any Weingarten function was located in Mathlib, physlib, or elsewhere (web searches, 11 June 2026). Caveat: absence from a finite search is not absence; re-verify on the Lean Zulip before any novelty claim appears in a publication.

Fact 3 (no canonical norm). Mathlib deliberately installs no norm instance on `MonoidAlgebra`: as with matrix norms, several natural choices exist. The ℓ^1 structure of Chapter 5 is therefore a *scoped* instance local to this project, following the `Matrix` precedent.

Attribution record (added 12 June 2026). The components of the parity theorem (Theorem 52) are displayed results: in the survey of Collins–Matsumoto–Novak (*The Weingarten Calculus*, Notices Amer. Math. Soc. **69** (2022), no. 5, 734–745; arXiv:2109.14890), Theorem 4.5 is the $1/N$ expansion with nonnegative weakly-monotone-walk coefficients and its parity constraint, Theorem 4.6 computes the leading coefficient as a strictly positive Catalan product, and Theorem 4.1 is the unique strictly monotone factorization (Jucys, via the Biane–Stanley labeling) — of which Lemma 50 is the existence half, strengthenable to uniqueness along the same induction. The expansion machinery is due to Novak (2010) and Matsumoto–Novak (*Jucys–Murphy elements and unitary matrix integrals*, Int. Math. Res. Not. IMRN **2013**, no. 2, 362–397; arXiv:0905.1992). No node of this blueprint is claimed as new mathematics; the Penrose-conditional route of Chapter 4 is a formalization-design choice. For the below-threshold package (12 June update): the Penrose characterization is the standard definition of the below-threshold Weingarten function (Collins–Śniady 2006; Zinn-Justin, arXiv:0907.2719); the group-algebra invertibility root set for *nonnegative* parameter, $\{0, \dots, n-1\}$, is stated as known by Collins–Matsumoto (arXiv:2510.21186, §3.2), where the sign-twist symmetry $\mathrm{Wg}(\sigma, -z) = (-1)^n \operatorname{sgn}(\sigma)\, \mathrm{Wg}(\sigma, z)$ also appears (invertible range) — for an *unrestricted* real or complex parameter z the singular set is the full integer band $\{-(n-1), \dots, n-1\}$ (the content spectrum $\prod_{(i,j)\in\lambda}(z + j - i)$; already at $n = 2$, $z = -1$ the element $\delta_e - \delta_{(0\,1)}$ is a zero divisor), so the positive-only set printed there *requires* the nonnegativity qualification made explicit here (as printed, for unrestricted z it conflicts with that paper's own display (3.2) and sign-twist symmetry); the same paper displays the rising sum rule (Theorem 34) as classical inside the proof of its Lemma 2.2 (present since v1, 24 October 2025) — a located display of known mathematics, altering no attribution here; the exact vanishing of the signed sum below threshold, the two-line evaluations through the Penrose equations, and the alternating-element witness remain unlocated — reference request outstanding.

The orthogonal and symplectic packages (Chapters 10–12) carry the same discipline. The absorption identities and the crossing-sign factorization are proved in full — informal proofs recorded in place, every step machine-checked in `scripts/` — and the conditional sum rules mirror Chapter 4. Above-threshold signed sums are in print (Campbell–Yin, arXiv:1907.01009;

the one-argument symplectic twist, arXiv:1412.0646, Redelmeier); the below-threshold vanishings, the eigenrelation form of the absorptions, and the expected-Pfaffian compression formula remain unlocated and are presumed routine — no novelty is inferred from absence (the full search ledger is maintained with the project's local records).

Standing conventions

$n \in \mathbb{N}$; the group is $S_n := \mathrm{Perm}(\mathrm{Fin}\, n)$ with 0-indexed points; N is a *real* parameter under the standing hypothesis $n - 1 < N$ where indicated. Every integer $N \geq n$ satisfies the hypothesis, so integer-N statements are special cases. For $\sigma \in S_n$, $c(\sigma)$ is the number of orbits of σ on $\mathrm{Fin}\, n$ (fixed points included) and $|\sigma| := n - c(\sigma)$. Products of possibly noncommuting factors are taken in the displayed index order.

Chapter 2

The group algebra and the dual Jucys–Murphy elements

Definition 1 (Group algebra)**.** $\mathcal{A}_n := \mathrm{MonoidAlgebra}(\mathbb{R}, S_n)$, the real group algebra of S_n, with basis $(\delta_\sigma)_{\sigma \in S_n}$ and convolution product $\delta_\sigma \delta_\tau = \delta_{\sigma\tau}$.

Lemma 2 (Mathlib: decomposition at 0)**.** *For each $\sigma \in S_{n+1}$ there exist unique $p \in \mathrm{Fin}(n{+}1)$ and $e \in S_n$ with $\sigma = (0\, p) \cdot \iota(e)$, where $\iota : S_n \hookrightarrow S_{n+1}$ fixes 0 and acts on successors; moreover $p = \sigma(0)$.*

Definition 3 (Jucys–Murphy elements, dual convention)**.** For $i \in \mathrm{Fin}\, n$,

$$K_i := \sum_{j>i} \delta_{(i\, j)} \in \mathcal{A}_n;$$

in particular $K_{n-1} = 0$ and K_0 has $n - 1$ terms.

Remark 4 (Dual versus textbook convention). The textbook elements are $J_i = \sum_{a<i} (a\, i)$. Conjugation by the order-reversing involution $x \mapsto n - 1 - x$ is an algebra automorphism carrying K_i to the (0-indexed) textbook J_{n-1-i}; it preserves the augmentation, the sign homomorphism, cycle counts, and the ℓ^1 norm, so every statement below is equivalent to its textbook form. The K convention matches `decomposeFin`'s split at 0 and removes an index-reversal layer from every induction; it is the single deliberate divergence from the standard textbook convention, and it is validated numerically by check C3 of `scripts/convention_checks.py`.

Lemma 5 (Commutativity)**.** *$K_i K_j = K_j K_i$ for all $i, j \in \mathrm{Fin}\, n$.*

Proof. Direct computation with transpositions, as in the classical case. This is used only to normalize factor order in Chapter 7; the factorization theorem itself fixes one order and never needs it. ☐

Chapter 3

Cycle count and the Gram element

Definition 6 (Cycle count)**.** $c(\sigma) :=$ (number of fixed points of σ) + (number of nontrivial cycles of σ), i.e. the number of orbits of σ on $\mathrm{Fin}\, n$.

Lemma 7 (Splice rule)**.** *With $\sigma = (0\,p) \cdot \iota(e) \in S_{n+1}$: if $p = 0$ then $c(\sigma) = c(e) + 1$, otherwise $c(\sigma) = c(e)$.*

Proof. If $p = 0$ then $\sigma = \iota(e)$ and 0 is a new fixed point. If $p \neq 0$, the point 0 is spliced into the orbit of p: one checks $\sigma(0) = p$, and $\sigma(x) = 0$ for the unique x with $\iota(e)(x) = p$, all other values unchanged; orbits not containing p are untouched and p's orbit absorbs 0, leaving the count unchanged. This is the hardest purely combinatorial lemma of the project; it is verified exhaustively for $n + 1 \leq 6$ by check C2 of `scripts/convention_checks.py`. □

Definition 8 (Gram element)**.** $\mathcal{G}_n(N) := \sum_{\sigma \in S_n} N^{c(\sigma)}\, \delta_\sigma \in \mathcal{A}_n$, for N in an arbitrary commutative coefficient ring k (the Lean declaration is stated at this generality; the real case of the ℓ^1 chapter is the specialization $k = \mathbb{R}$).

Theorem 9 (Jucys factorization, dual form)**.** *For every N in an arbitrary commutative coefficient ring k (the Lean theorem is stated at this generality; all real, rational, and complex uses below are specializations):*

$$\mathcal{G}_n(N) = \prod_{i=0}^{n-1} (N + K_i),$$

the product taken in increasing index order.

Proof. Induction on n; the base $n = 0$ reads $1 = 1$. For the step, decompose $\sigma = (0\,p)\,\iota(e)$ by Lemma 2 and apply Lemma 7:

$$\mathcal{G}_{n+1}(N) = \sum_{e \in S_n} \Big(N^{c(e)+1}\, \iota(e) + \sum_{p \neq 0} N^{c(e)}\, (0\,p)\, \iota(e) \Big) = (N + K_0)\, \iota_*(\mathcal{G}_n(N)),$$

where ι_* is the algebra map induced by ι. Since ι_* carries $K_i^{(n)}$ to $K_{i+1}^{(n+1)}$, the inductive hypothesis turns the right factor into $\prod_{i=1}^{n} (N + K_i^{(n+1)})$. Check C3 of `scripts/convention_checks.py` verifies the identity exactly for $n \leq 5$ at several rational N. □

Chapter 4

The below-threshold package: Penrose-conditional sum rules

The two sum rules are evaluations of one-dimensional characters, and those evaluations pass through the *Penrose equations* $\mathcal{G}\mathcal{W}\mathcal{G} = \mathcal{G}$, $\mathcal{W}\mathcal{G}\mathcal{W} = \mathcal{W}$ exactly as they pass through inverses. Stating the rules conditionally on a Penrose partner makes them valid at *every* N — including below threshold, where the Gram element is singular — with no irreducible-character theory, no Schur functions and no spectral theory (the two one-dimensional characters, augmentation and sign, are of course exactly what the sum rules evaluate). The stable-range rules of Chapter 6 become one-line corollaries, so the analytic chapter is needed only for existence of the inverse and for the parity theorem. This is a formalization-design choice, not a claim of new theorems. Indeed the characterization itself is the literature's standard definition of the below-threshold Weingarten function: following Collins–Śniady (*Comm. Math. Phys.* **264** (2006), 773–795), Wg is conventionally the pseudo-inverse of the Gram data; P. Zinn-Justin (arXiv:0907.2719) records the convention in this two-equation form with a symmetry qualifier (there, Wg is the *symmetric* matrix satisfying the two equations), and the bare two-equation form appears in arXiv:0906.4694 (before its Theorem 2.1). The two-line character evaluations below, their conditional formulation, and the alternating-element witness have not been located in print (searches of 11–12 June 2026; reference request outstanding) and are presumed routine to experts. Throughout, the coefficient ring is an arbitrary commutative ring k (a field where inverses are displayed); the Lean development is generalized accordingly, with only the ℓ^1 and parity chapters pinned to $\mathbb{R}$. Checks B1–B5 of `scripts/convention_checks.py` pin every identity below in exact rational arithmetic.

Lemma 10 (Coefficient formulas for the two characters)**.** *Via `MonoidAlgebra.lift`, the trivial character and the sign character extend to unital algebra homomorphisms $\varepsilon, \hat{s} : \mathcal{A}_n \to \mathbb{R}$ with $\varepsilon(x) = \sum_\sigma x_\sigma$ and $\hat{s}(x) = \sum_\sigma \operatorname{sgn}(\sigma)\, x_\sigma$. (Lean names: `Weingarten.aug_apply`, `Weingarten.sgnHom_apply`.)*

Lemma 11 (Values on K_i)**.** *$\varepsilon(K_i) = n - 1 - i$ and $\hat{s}(K_i) = -(n-1-i)$. (Lean names: `Weingarten.aug_jm`, `Weingarten.sgn_jm`; verified numerically by check C4.)*

Lemma 12 (Character values of the Gram element, all N)**.** *For every $N \in k$: $\varepsilon(\mathcal{G}_n(N)) = \prod_{m=0}^{n-1}(N+m)$ and $\hat{s}(\mathcal{G}_n(N)) = \prod_{m=0}^{n-1}(N-m)$. (Lean names: `Weingarten.aug_gram`, `Weingarten.sgnHom_gram`; check B2, including $N < n$ and fractional N.)*

Proof. The Jucys factorization is a polynomial identity valid for all N with no invertibility anywhere; apply each homomorphism factorwise and reindex by $m = n - 1 - i$. $\square$

Definition 13 (Alternating element). $a_n := \sum_\sigma \operatorname{sgn}(\sigma)\, \delta_\sigma \in \mathcal{A}_n$.

Lemma 14 (Absorption). *$a_n\, x = \hat{s}(x)\, a_n$ for every $x \in \mathcal{A}_n$ (check B1).*

Proof. By linearity it suffices that $a_n\, \delta_g = \operatorname{sgn}(g)\, a_n$, which is the substitution $\rho = \sigma g$ together with multiplicativity of sgn. $\square$

Definition 15 (All-ones element). $s_n := \sum_{\sigma \in S_n} \delta_\sigma \in \mathcal{A}_n$ — the augmentation twin of Definition 13.

Lemma 16 (Ones absorption). *$s_n\, x = \varepsilon(x)\, s_n$ for every $x \in \mathcal{A}_n$: right multiplication by a basis element only permutes the summands.*

Proof. Mirror of Lemma 14 without signs: reindex by right translation, then extend by linearity. $\square$

Theorem 17 (Gram singularity below threshold). *Over $\mathbb{Q}$, for every integer $1 \le N < n$ the Gram element $\mathcal{G}_n(N)$ is not a unit. The statement is known: for* nonnegative *parameter, invertibility of $\mathcal{G}_n(N)$ in the group algebra holds exactly off the integer root set $\{0, 1, \dots, n-1\}$ (stated as known by Collins–Matsumoto, arXiv:2510.21186, §3.2). For an* unrestricted *real or complex parameter z the singular set is the full integer band $\{-(n-1), \dots, n-1\}$: the central character of $\mathcal{G}_n(z)$ on the irreducible indexed by $\lambda \vdash n$ is the content product $\prod_{(i,j)\in\lambda}(z + j - i)$, whose roots sweep the band (already at $n = 2$, $z = -1$ the element $\mathcal{G}_2(-1) = \delta_e - \delta_{(0\,1)}$ is a zero divisor); the positive-only set printed in arXiv:2510.21186 therefore* requires *the nonnegativity qualification adopted here — as printed, for unrestricted z it conflicts with the paper's own display (3.2) and sign-twist symmetry, which confirm the band. The formalized content of this node is the singular direction for natural $1 \le N < n$; the negative-integer companion is Theorem 18, and the invertible direction off the band (a character-spectrum fact) is* not *formalized in this repository — the Lean development proves invertibility only on the real range $N > n - 1$ (Definition 30). The alternating-element proof below is the part we have not located in print.*

Proof. $a_n\, \mathcal{G}_n(N) = \hat{s}(\mathcal{G}_n(N))\, a_n = \left(\prod_{m<n}(N - m)\right) a_n = 0$, the factor $m = N$ vanishing, while $a_n \neq 0$ in characteristic zero; a unit cannot be a right zero divisor. (Checks B3, B3b.) $\square$

Theorem 18 (Gram singularity at negative integers). *For integers $0 \le m \le n - 1$ the Gram element at $N = -m$ is not a unit: $s_n\, \mathcal{G}_n(-m) = \left(\prod_{j<n}(-m + j)\right) s_n = 0$ while $s_n \neq 0$ in characteristic zero. With Theorem 17 this exhibits the full integer singular band $\{-(n-1), \dots, n-1\}$: the alternating element detects the positive-side roots through the signed character, the all-ones element detects the negative-side roots through the augmentation.*

Proof. The rising factorial (Lemma 12) vanishes at $-m$ through the factor $j = m$; s_n is a zero divisor, so $\mathcal{G}_n(-m)$ is not invertible. (Machine check: the $n = 2$ determinant band scan in `verify_penrose_eigen.py`.) $\square$

Definition 19 (Penrose pair). $\mathcal{W}$ is a *Penrose partner* of $\mathcal{G}$ if $\mathcal{G}\mathcal{W}\mathcal{G} = \mathcal{G}$ and $\mathcal{W}\mathcal{G}\mathcal{W} = \mathcal{W}$. Any Moore–Penrose pseudo-inverse qualifies; so does the genuine inverse when it exists. In a general matrix algebra these two equations define a *reflexive generalized inverse* and do not pin down a unique $\mathcal{W}$; for the *central* Gram element they do (Theorem 22 below), so the two-equation characterization is canonical here whenever it is inhabited. This pair of equations is the literature's standard convention for the below-threshold Weingarten function (Collins–Śniady

2006; recorded in this two-equation form by Zinn-Justin, arXiv:0907.2719, and in the orthogonal setting in *The orthogonal Weingarten formula in compact form*, arXiv:0906.4694).

Lemma 20 (Centrality of the Gram element)**.** *$\mathcal{G}_n(N)$ commutes with every element of the group algebra: $N^{c(\sigma)}$ is a class function, so both products against a basis element reindex to the same sum.*

Proof. $c(\rho g^{-1}) = c(g^{-1}\rho)$ by conjugation invariance of the cycle count; extend by linearity. □

Lemma 21 (Group-inverse uniqueness)**.** *In any monoid, two Penrose partners of g that both commute with g coincide.*

Proof. $w_1 = w_1^2 g$ and $w_2 = g w_2^2$ from the second Penrose equations plus commutation; inserting $g = g w_2 g$ and $g = g w_1 g$ gives $w_1 = w_1 w_2 g = w_2$. Associativity only. □

Theorem 22 (Canonicity of the Gram's Penrose partner)**.** *Any two Penrose partners of $\mathcal{G}_n(N)$ coincide: the commutation hypothesis is free by centrality (Lemma 20), so the two-equation characterization of the below-threshold Weingarten data is canonical whenever it is inhabited.*

Proof. Immediate from Lemmas 21 and 20. □

Theorem 23 (Conditional rising rule, all N)**.** *Over a field, if $\mathcal{W}$ is a Penrose partner of $\mathcal{G}_n(N)$ and $r := \prod_{m=0}^{n-1}(N+m) \neq 0$ — true for every real $N > 0$, in particular every positive integer including $N < n$ — then $\varepsilon(\mathcal{W}) = r^{-1}$.*

Proof. Apply ε to $\mathcal{G}\mathcal{W}\mathcal{G} = \mathcal{G}$: $r^2\,\varepsilon(\mathcal{W}) = r$. □

Theorem 24 (Conditional falling rule off the root set)**.** *Over a field, if $\mathcal{W}$ is a Penrose partner of $\mathcal{G}_n(N)$ and $f := \prod_{m=0}^{n-1}(N-m) \neq 0$, then $\hat{s}(\mathcal{W}) = f^{-1}$. This recovers the stable-range falling rule and extends it to every N off the root set $\{0, 1, \dots, n-1\}$.*

Proof. Apply $\hat{s}$ to $\mathcal{G}\mathcal{W}\mathcal{G} = \mathcal{G}$: $f^2\,\hat{s}(\mathcal{W}) = f$. □

Theorem 25 (Exact vanishing below threshold)**.** *Over any commutative ring, if $\mathcal{W}\mathcal{G}_n(N)\mathcal{W} = \mathcal{W}$ and $\prod_{m=0}^{n-1}(N-m) = 0$ — e.g. every integer $1 \le N < n$ — then $\hat{s}(\mathcal{W}) = 0$.*

Proof. $\hat{s}(\mathcal{W}) = \hat{s}(\mathcal{W})\,\hat{s}(\mathcal{G})\,\hat{s}(\mathcal{W}) = \hat{s}(\mathcal{W}) \cdot 0 \cdot \hat{s}(\mathcal{W}) = 0$. □

Theorem 26 (Forced breakdown of the sign pattern)**.** *Over $\mathbb{R}$, under the hypotheses of Theorem 25, the strict parity pattern $0 < (-1)^{n-c(\sigma)}\,\mathcal{W}_\sigma$ cannot hold for every σ.*

Proof. A nonempty sum of strictly positive terms is positive, contradicting $\hat{s}(\mathcal{W}) = 0$. □

Chapter 5

The scoped ℓ^1 structure and invertibility

Definition 27 (Scoped ℓ^1 norm)**.** $\|x\|_1 := \sum_\sigma |x_\sigma|$ on $\mathcal{A}_n$. This is a complete algebra norm: submultiplicativity is the finite Young inequality for convolution on a group. Installed as a *scoped* `NormedRing` instance (Fact 3 of Chapter 1).

Lemma 28 (Norm of K_i)**.** $\|K_i\|_1 = n - 1 - i \leq n - 1$.

Lemma 29 (Invertibility)**.** *If $n - 1 < N$ then each $N + K_i$ is a unit of $\mathcal{A}_n$, with inverse the absolutely convergent Neumann series $\sum_{k \geq 0} (-1)^k K_i^k N^{-k-1}$.*

Proof. $\|K_i/N\|_1 \leq (n-1)/N < 1$; apply the geometric-series construction of units in a complete normed ring (Mathlib's `Units.oneSub` family). □

Chapter 6

The Weingarten element and the two sum rules

Definition 30 (Weingarten element)**.** For $n-1<N$ the element $\mathcal{G}_n(N)$ is a unit, being a product of units by Theorem 9 and Lemma 29; set $\mathcal{W}_n(N) := \mathcal{G}_n(N)^{-1} \in \mathcal{A}_n$.

Definition 31 (Weingarten function)**.** $\mathrm{Wg}(\sigma, N) :=$ the δ_σ-coefficient of $\mathcal{W}_n(N)$.

Theorem 32 (Inverse property)**.** *$\mathcal{G}_n(N)\,\mathcal{W}_n(N) = 1 = \mathcal{W}_n(N)\,\mathcal{G}_n(N)$ for $n-1<N$.*

Proof. Immediate: $\mathcal{W}_n(N)$ is the inverse of a unit. □

Lemma 33 (The stable inverse is a Penrose partner)**.** *For $n-1<N$ the Weingarten element $\mathcal{W}_n(N)$ is a Penrose partner of $\mathcal{G}_n(N)$: both equations follow from $\mathcal{G}\mathcal{W} = 1 = \mathcal{W}\mathcal{G}$.*

Theorem 34 (Rising sum rule)**.** *For $n-1<N$:*

$$\sum_{\sigma \in S_n} \mathrm{Wg}(\sigma, N) = \prod_{m=0}^{n-1} \frac{1}{N+m}.$$

Proof. Instantiate Theorem 23 at the Penrose partner of Lemma 33; $\prod_m (N+m) > 0$ in the stable range, and $\varepsilon(\mathcal{W})$ is the coefficient sum by Lemma 10. □

Theorem 35 (Falling sum rule)**.** *For $n-1<N$:*

$$\sum_{\sigma \in S_n} \mathrm{sgn}(\sigma)\ \mathrm{Wg}(\sigma, N) = \prod_{m=0}^{n-1} \frac{1}{N-m}.$$

Proof. Instantiate Theorem 24 likewise; $\prod_m (N-m) > 0$ since $N > n-1$. □

Remark 36 (Below threshold: in and out of scope)*.* The all-N statements are now *in* scope in conditional form (Chapter 4): for any Penrose partner the rising value persists at every $N>0$ and the signed sum vanishes exactly for integer $1 \le N < n$. What remains out of scope is the *general existence* of a Penrose partner below threshold and its identification with the restricted character-sum definition (Chapter 14); the cell of Chapter 9 is the certified existence instance.

Chapter 7

The parity sign theorem

Lemma 37 (Sign of a word)**.** *The sign of a product of k transpositions is $(-1)^k$; consequently any word of transpositions with product σ has length $\equiv |\sigma| \pmod 2$. Proved via Mathlib's `Equiv.Perm.sign_prod_list_swap`.*

The next six lemmas are the verified analytic engine of Lemma 49: they reduce $\mathcal{W}_n(N)$, in the stable range, to a single absolutely convergent sum over slot multi-indices. The remaining gap in Lemma 49 is purely the combinatorial coefficient-extraction from that sum.

Lemma 38 (Single-factor Neumann series)**.** *For $n-1 < N$ and each slot i,*

$$(N + K_i)^{-1} = N^{-1} \sum_{k \geq 0} \left(-N^{-1} K_i\right)^k,$$

the series converging absolutely since $\|N^{-1}K_i\|_1 < 1$.

Proof. Write $N + K_i = N \cdot (1-x)$ with $x = -N^{-1}K_i$; then $(N+K_i)^{-1} = N^{-1}(1-x)^{-1}$ and $(1-x)^{-1} = \sum_k x^k$ by the geometric series in a Banach algebra (Mathlib `geom_series_eq_inverse`). □

Lemma 39 (Inverse of a commuting product)**.** *For a list of pairwise-commuting units in any ring, $\left(\prod_i a_i\right)^{-1} = \prod_i a_i^{-1}$ in the* same *order: the anti-homomorphism reversal of $(-)^{-1}$ is undone by commutativity. A general ring lemma, reused below.*

Lemma 40 (Product form of $\mathcal{W}$)**.** *For $n-1 < N$, $\mathcal{W}_n(N) = \displaystyle\prod_{i=0}^{n-1} (N+K_i)^{-1}$, in increasing slot order.*

Proof. Invert the Jucys factorization $\mathcal{G}_n(N) = \prod_i (N + K_i)$ (Theorem 9) factorwise: the $N + K_i$ are units (Lemma 29) and pairwise commute (Lemma 5, with N central), so Lemma 39 applies. □

Lemma 41 (Product of Neumann series)**.** *For $n-1 < N$, $\mathcal{W}_n(N) = \displaystyle\prod_{i=0}^{n-1} \Big(N^{-1} \sum_{k\geq 0} (-N^{-1}K_i)^k \Big)$.*

Proof. Expand each factor of Lemma 40 by the single-factor series of Lemma 38. □

Lemma 42 (n-fold Cauchy product)**.** *In a complete normed ring, an ordered product of absolutely convergent series distributes into a single sum over multi-indices, preserving the order:*

$$\prod_{i=0}^{n-1}\Big(\sum_{k\geq 0} f_i(k)\Big) = \sum_{\kappa\colon \mathrm{Fin}\, n\to\mathbb{N}} \prod_{i=0}^{n-1} f_i(\kappa_i).$$

The summability side-condition — that the multi-indexed product is absolutely summable — is `Weingarten.summable_list_prod_norm`*.*

Proof. Induction on n, applying the binary Cauchy product (Mathlib `tsum_mul_tsum_of_summable_norm`) at each step and reindexing $\mathbb{N}\times(\mathrm{Fin}\, n\to\mathbb{N})\simeq(\mathrm{Fin}(n{+}1)\to\mathbb{N})$ via `Fin.consEquiv`. □

Lemma 43 (Multi-index form of $\mathcal{W}$)**.** *For $n-1<N$,*

$$\mathcal{W}_n(N) = \sum_{\kappa\colon \mathrm{Fin}\, n\to\mathbb{N}} \prod_{i=0}^{n-1}\Big(N^{-1}\,(-N^{-1}K_i)^{\kappa_i}\Big).$$

Proof. Rewrite each factor of Lemma 41 as $\sum_k N^{-1}(-N^{-1}K_i)^k$ and apply the n-fold Cauchy product (Lemma 42); absolute summability per slot is the geometric bound $\|N^{-1}K_i\|_1<1$. □

Lemma 44 (Scalar/word separation)**.** *For $n-1<N$, writing $|\kappa| := \sum_i \kappa_i$,*

$$\mathcal{W}_n(N) = \sum_{\kappa\colon \mathrm{Fin}\, n\to\mathbb{N}} (-1)^{|\kappa|}\, N^{-(n+|\kappa|)} \prod_{i=0}^{n-1} K_i^{\kappa_i}.$$

Each term is an explicit scalar — depending only on the total degree $|\kappa|$ — times the swap-word product $\prod_i K_i^{\kappa_i}$, a nonnegative-integer combination of permutations.

Proof. Pull the scalars out of each factor of Lemma 43: $N^{-1}(-N^{-1}K_i)^{\kappa_i} = (-1)^{\kappa_i}N^{-(\kappa_i+1)}K_i^{\kappa_i}$, and the exponents sum as $\sum_i(\kappa_i+1) = n+|\kappa|$. □

Lemma 45 (Coordinate-coefficient linearization)**.** *For $n-1<N$ and each σ, with $|\kappa| := \sum_i \kappa_i$,*

$$\mathrm{Wg}(\sigma,N) = \sum_{\kappa\colon \mathrm{Fin}\, n\to\mathbb{N}} (-1)^{|\kappa|}\, N^{-(n+|\kappa|)}\, (\textstyle\prod_i K_i^{\kappa_i})(\sigma),$$

where $(\prod_i K_i^{\kappa_i})(\sigma)$ is the δ_σ-coefficient of the swap-word product (a nonnegative integer). A real-valued series.

Proof. Apply the δ_σ-coordinate functional $x\mapsto x(\sigma)$ to Lemma 44. The functional is continuous — indeed $\mathcal{A}_n = \mathbb{R}[S_n]$ is finite-dimensional, so every linear functional is continuous — hence commutes with the absolutely convergent sum, and it scales each term: $((-1)^{|\kappa|}N^{-(n+|\kappa|)}\prod_i K_i^{\kappa_i})(\sigma) = (-1)^{|\kappa|}N^{-(n+|\kappa|)}(\prod_i K_i^{\kappa_i})(\sigma)$. □

Lemma 46 (Integral coefficients)**.** *For $n-1<N$ and each σ, with $|\kappa| := \sum_i \kappa_i$,*

$$\mathrm{Wg}(\sigma,N) = \sum_{\kappa\colon \mathrm{Fin}\, n\to\mathbb{N}} (-1)^{|\kappa|}\, N^{-(n+|\kappa|)}\, c_\kappa(\sigma),$$

where $c_\kappa(\sigma)\in\mathbb{N}$ is the δ_σ-coefficient of the swap-word product $\prod_i K_i^{\kappa_i}$ taken over $\mathbb{N}$ — the number of factorizations of σ as an ordered product of κ_i slot-i transpositions.

Proof. The coefficient $(\prod_i K_i^{\kappa_i})(\sigma)$ of Lemma 45 is the image, under the coefficient-cast ring homomorphism $\mathbb{N}[S_n] \to \mathbb{R}[S_n]$, of the same word formed from the $\mathbb{N}$-coefficient elements $K_i^{\mathbb{N}} = \sum_{j>i}(i\,j)$; the ring homomorphism acts coefficient-wise, so the real coefficient is the Nat.cast of the integer count $c_\kappa(\sigma)$. ☐

Lemma 47 (Degree regrouping)**.** *For $n-1 < N$ and each σ,*

$$\mathrm{Wg}(\sigma, N) = \sum_{k\geq 0}(-1)^k\, N^{-(n+k)}\, m_k(\sigma), \qquad m_k(\sigma) := \sum_{|\kappa|=k} c_\kappa(\sigma) \in \mathbb{N},$$

the sum over κ of degree k taken over the finite set of slot-degree tuples (the antidiagonal tuple antidiagonalTuple $n\,k$*).*

Proof. The series of Lemma 46 is absolutely convergent (the δ_σ-coordinate of the summable word product), so it may be regrouped by the total degree $|\kappa| = \sum_i \kappa_i$. The fibers of the degree map are the finite tuple-antidiagonals, and the scalar $(-1)^{|\kappa|}N^{-(n+|\kappa|)}$ is constant on each fiber, factoring out of the (finite) inner sum that defines $m_k(\sigma)$. ☐

Lemma 48 (Parity restriction)**.** *For each σ and each k, if $m_k(\sigma) \neq 0$ then $k \equiv |\sigma| \pmod 2$, where $|\sigma| = n - c(\sigma)$.*

Proof. A nonzero count $m_k(\sigma)$ exhibits a slot-degree tuple κ of degree k with $(\prod_i K_i^{\kappa_i})(\sigma) \neq 0$. Each K_i is supported on transpositions (sign -1), and support signs multiply, so the swap-word product $\prod_i K_i^{\kappa_i}$ is supported only on permutations of sign $(-1)^{|\kappa|} = (-1)^k$ (formalized as a sign-homogeneity invariant, `Weingarten.word_support_sign`). Hence $\mathrm{sign}(\sigma) = (-1)^k$; comparing with $\mathrm{sign}(\sigma) = (-1)^{n-c(\sigma)}$ (`Weingarten.sign_eq_neg_one_pow`) forces $k \equiv n - c(\sigma) \pmod 2$. ☐

Lemma 49 (Nonnegative expansion)**.** *For $n-1 < N$ and each σ,*

$$(-1)^{|\sigma|}\ \mathrm{Wg}(\sigma, N) = \sum_{k\geq 0} m_k(\sigma)\, N^{-(n+k)},$$

where $m_k(\sigma) \in \mathbb{Z}_{\geq 0}$ counts the tuples of words $(w_0, \dots, w_{n-1})$ — each w_i a word in the transpositions $\{(i\,j) : j > i\}$ supporting K_i — of total length k whose ordered product is σ; and $m_k(\sigma) = 0$ unless $k \equiv |\sigma| \pmod 2$.

Proof. Lemma 47 expresses $\mathrm{Wg}(\sigma, N) = \sum_{k\geq 0}(-1)^k N^{-(n+k)} m_k(\sigma)$ with nonnegative integer counts $m_k(\sigma) \in \mathbb{N}$, and Lemma 48 gives $m_k(\sigma) = 0$ unless $k \equiv |\sigma| \pmod 2$. On every surviving term $k \equiv |\sigma|$, so $(-1)^k = (-1)^{|\sigma|}$; multiplying the absolutely convergent series through by $(-1)^{|\sigma|}$ (as a `HasSum`) collapses the alternating sign and gives the stated nonnegative expansion. The minimal term $m_{|\sigma|}(\sigma) \neq 0$ is Lemma 51. ☐

Lemma 50 (A shortest slot-word exists)**.** *For every $\sigma \in S_n$ there is a tuple $(w_0, \dots, w_{n-1})$ in which each w_i is either empty or a single transposition $(i\,j_i)$ with $j_i > i$, whose ordered product is σ and whose total length is exactly $|\sigma| = n - c(\sigma)$. Hence $m_{|\sigma|}(\sigma) \geq 1$.*

Proof. A *slot-word tuple* is a tuple $(w_0, \dots, w_{n-1})$ in which each w_i is either empty (denoting id) or a single slot-i transposition $(i\,j_i)$ with $j_i > i$; its product is $\pi(w) := w_0 w_1 \cdots w_{n-1}$ taken in increasing slot order, and its length $\ell(w)$ is the number of nonempty slots. We must produce, for each σ, such a tuple with $\pi(w) = \sigma$ and $\ell(w) = |\sigma| = n - c(\sigma)$. We induct on n.

Length step. First record the one quantitative input. If $\sigma \in S_{n+1}$ decomposes as $\sigma = (0\,p)\,\iota(e)$ via Lemma 2 (with $p = \sigma(0)$, $e \in S_n$, and ι the embedding fixing 0 and sending $x \mapsto x+1$ on successors), then the splice rule (Lemma 7) gives $c(\sigma) = c(e) + 1$ when $p = 0$ and $c(\sigma) = c(e)$ when $p \neq 0$. Since $|\sigma| = (n+1) - c(\sigma)$ and $|e| = n - c(e)$, this reads $|\sigma| = |e|$ for $p = 0$ and $|\sigma| = |e| + 1$ for $p \neq 0$: a genuine slot-0 transposition raises $|\cdot|$ by exactly one, and the empty slot 0 leaves it unchanged.

Base case $n = 0$. S_0 is trivial, so $\sigma = \mathrm{id}$; the empty tuple is the unique slot-word tuple, with product $\mathrm{id} = \sigma$ and length 0. As $c(\mathrm{id}) = 0$ we have $|\sigma| = 0$, so $\ell(w) = 0 = |\sigma|$.

Inductive step. Assume the statement for S_n and take $\sigma \in S_{n+1}$. Apply Lemma 2 to get $p = \sigma(0)$ and $e \in S_n$ with $\sigma = (0\,p)\,\iota(e)$, and let $v = (v_0, \dots, v_{n-1})$ be the slot-word tuple for e supplied by the inductive hypothesis, with $\pi(v) = e$ and $\ell(v) = |e|$. Define $w = (w_0, \dots, w_n)$ by

$$w_0 := \begin{cases} \text{empty}, & p = 0, \\ (0\,p), & p \neq 0, \end{cases} \qquad w_{i+1} := \iota(v_i) \quad (i \in \mathrm{Fin}\, n).$$

Because ι fixes 0 and sends $x \mapsto x{+}1$ on successors, it carries a slot-i transposition $(i\,j)$ $(j > i)$ to $(i{+}1\, j{+}1)$ with $j{+}1 > i{+}1$, and it carries the empty entry to the empty entry; hence each w_{i+1} is a legal slot-$(i{+}1)$ entry, slot 0 is set independently, and no slot is reused, so w is a bona fide slot-word tuple for S_{n+1}.

Since ι is a homomorphism it preserves ordered products, so $w_1 \cdots w_n = \iota(v_0) \cdots \iota(v_{n-1}) = \iota(e)$, and prepending slot 0 gives $\pi(w) = w_0\,\iota(e)$. When $p = 0$, $w_0 = \mathrm{id} = (0\,0) = (0\,p)$; when $p \neq 0$, $w_0 = (0\,p)$. In both cases $\pi(w) = (0\,p)\,\iota(e) = \sigma$. For the length, the shift preserves emptiness, so the block $w_1, \dots, w_n$ has exactly $\ell(v) = |e|$ nonempty slots, while slot 0 contributes 0 if $p = 0$ and 1 if $p \neq 0$. By the length step the resulting total $\ell(w)$ equals $|\sigma|$ in both branches, completing the induction.

Finally, the constructed tuple is one factorization of σ of length $|\sigma|$, so it contributes a term ≥ 1 to $m_{|\sigma|}(\sigma) = \sum_{|\kappa|=|\sigma|} c_\kappa(\sigma)$ (taking $\kappa_i = 1$ on the nonempty slots); hence $m_{|\sigma|}(\sigma) \geq 1$, the nonvanishing recorded in Lemma 51. Verified constructively for all σ with $n \leq 5$ by check C5.

Reduced form. `exists_min_word` is the same induction on n run directly on the witness data $f : \mathrm{Fin}\, n \to \mathrm{Option}\,(\mathrm{Fin}\, n)$. The `zero` case returns the empty function (`i.elim0`) and discharges the trivial product by `Subsingleton.elim`; the `succ` case sets $(p, e) :=$ `decomposeFin` σ, recurses to get f' for e, and defines $f :=$ `Fin.cases` (if $p = 0$ then none else some p) $(\lambda j,\ (f'\ j)$.`map Fin.succ`). The three bundled goals are then closed mechanically: monotonicity via `Fin.succ_lt_succ_iff`, the product via `iotaHom_swap` together with `map_list_prod (iotaHom n)` and `decomposeFin_symm_eq`, and the length count via `cycleCount_decomposeFin` closed by `omega`. ☐

Lemma 51 (Nonvanishing minimal term)**.** *$m_{|\sigma|}(\sigma) \neq 0$, i.e. the count at the minimal degree $k = |\sigma| = n - c(\sigma)$ is nonzero.*

Proof. The shortest slot-word of Lemma 50 is one factorization of σ of length $|\sigma|$, so it contributes a term ≥ 1 to the count $m_{|\sigma|}(\sigma) = \sum_{|\kappa|=|\sigma|} c_\kappa(\sigma)$. (Formally: in $\mathbb{N}[S_n]$ a single factorization term lower-bounds the convolution coefficient, $\prod_i (K_i^{\mathbb{N}})^{\kappa_i}$ evaluated at σ is at least the product of the chosen basis coefficients, all equal to 1.) ☐

Theorem 52 (Parity sign theorem)**.** *For $n - 1 < N$ and every $\sigma \in S_n$:*

$$(-1)^{|\sigma|}\, \mathrm{Wg}(\sigma, N) > 0;$$

in particular $\mathrm{Wg}(\sigma, N) \neq 0$ *and its sign is* $\mathrm{sgn}(\sigma)$.

Proof. The expansion has nonnegative terms, and the term $k = |\sigma|$ is at least $N^{-(n+|\sigma|)} > 0$. ☐

Chapter 8

The cancellation ratio in closed form

Theorem 53 (Closed form)**.** *For $n-1<N$:*

$$\sum_\sigma \left|\mathrm{Wg}(\sigma,N)\right| = \prod_{m=0}^{n-1}\frac{1}{N-m}, \qquad \frac{\sum_\sigma \mathrm{Wg}(\sigma,N)}{\sum_\sigma \left|\mathrm{Wg}(\sigma,N)\right|} = \prod_{j=0}^{n-1}\frac{N-j}{N+j}.$$

(Lean names: `Weingarten.sum_abs_wg`, `Weingarten.cancellation_ratio`.)

Proof. By Theorem 52, $|\,\mathrm{Wg}(\sigma,N)| = \mathrm{sgn}(\sigma)\ \mathrm{Wg}(\sigma,N)$ (Lean: `Weingarten.abs_wg_eq`); apply the two sum rules and divide. □

Chapter 9

The certified cell $(N, n) = (2, 3)$

This chapter certifies, as a finite exact computation over $\mathbb{Q}$, the one below-threshold cell the development commits to — with no general pseudo-inverse theory and no analysis.

Definition 54 (Gram data)**.** $G : S_3 \times S_3 \to \mathbb{Q}$, $G(\sigma, \tau) := 2^{c(\sigma^{-1}\tau)}$, viewed as a 6×6 matrix over $\mathbb{Q}$.

Definition 55 (Candidate Weingarten data)**.** $W(\sigma, \tau) := w(\sigma^{-1}\tau)$, where $w(\mathrm{id}) = 17/144$, $w = 1/144$ on the three transpositions, and $w = -7/144$ on the two 3-cycles.

Lemma 56 (Penrose uniqueness)**.** *Over $\mathbb{Q}$, a matrix W satisfying $GWG = G$, $WGW = W$, $(GW)^\mathsf{T} = GW$, and $(WG)^\mathsf{T} = WG$ is unique — the standard six-line algebraic argument, valid over any commutative ring.*

Theorem 57 (The cell, certified)**.** *The pair (G, W) satisfies the four Penrose identities; by Lemma 56, W is therefore* the *pseudo-inverse of G, so the three values of Definition 55 are the $(N, n) = (2, 3)$ Weingarten values. Moreover every row of W sums to $1/24$ and every sign-weighted row sums to 0 — the below-threshold degeneration of the falling rule. Proof route: the data is integer-valued up to a global $1/144$, so the Penrose equations become the $\mathbb{Z}$-matrix identities $G W_0 G = 144 G$ etc., which kernel `decide` reduces (the rational `decide` stalls on `Rat` gcd-normalization); casting back through $\mathbb{Z} \to \mathbb{Q}$ recovers the statements. No `native_decide`, so the trusted base is unchanged. (Lean names: `Weingarten.Cell23.penrose`, `Weingarten.Cell23.row_sums`; verified exactly by check C6.)*

Proof. Finite exact computation plus Lemma 56. □

Definition 58 (Algebra-level cell data)**.** $\mathcal{W}_{2,3} := \sum_\sigma w(\sigma)\, \delta_\sigma \in \mathcal{A}_3$ over $\mathbb{Q}$, with w the candidate values of Definition 55.

Theorem 59 (Algebra-level Penrose certificate)**.** *$\mathcal{W}_{2,3}$ is a Penrose partner of $\mathcal{G}_3(2)$ over $\mathbb{Q}$. Proof: transport the matrix certificate of Theorem 57 along the injective left-regular embedding Φ = `toMatrixAlgEquiv` $\circ$ `Algebra.lmul`, whose matrix of $\mathcal{G}_3(2)$ is G and of $\mathcal{W}_{2,3}$ is W, using that w and N^c are inverse-symmetric class functions (`cycleCount_mul_inv_comm`, `wval_mul_inv_comm`).*

Corollary 60 (Cell sum rules by two routes)**.** *$\varepsilon(\mathcal{W}_{2,3}) = 1/24$ and $\hat{s}(\mathcal{W}_{2,3}) = 0$, as corollaries of the conditional rules; the `decide`-route row sums of Theorem 57 remain as an independent cross-check (check B4).*

Proposition 61 (Average-sign anchor)**.** $(\sum_\sigma w(\sigma))/(\sum_\sigma |w(\sigma)|) = 3/17$ *— the first below-threshold average-sign anchor, certified (check B4).*

Proposition 62 (Sign-flip witness)**.** *The transposition* $(0\,1)$ *has* $\mathrm{sgn} = -1$ *and* $w > 0$*: the concrete instance of the forced breakdown.*

Corollary 63 (Cell Gram singularity)**.** $\mathcal{G}_3(2)$ *is not a unit (instance of Theorem 17; numerically* $\det G = 0$*, check B3b).*

Corollary 64 (Uniqueness characterization)**.** *Any matrix satisfying the four transpose-form Penrose equations for* G *equals* W*: the certified values are* the $(2, 3)$ *Weingarten data.*

Chapter 10

Pair partitions: loops, crossings, and the twisted vectors

This chapter and the two that follow develop the orthogonal and symplectic analogues of the below-threshold package, fully proved in Lean: every declaration carries `leanok` and is *sorry*-free and axiom-clean, the informal proofs are recorded here in full, and every identity is pinned in exact rational arithmetic by the suites in `scripts/` (BO1–BO4, PL1–PL8, A1–A8, PS1–PS6, PF0–PF4). The complete prior-art ledger is maintained with the project's local records; the attribution paragraphs below summarize it. As everywhere in this blueprint, absence from our searches is *not* a novelty claim — reference requests are outstanding.

Pair partitions of $\{0, \dots, 2n-1\}$ are encoded as fixed-point-free involutions, written $\mathcal{P}_2(2n)$; the unitary chapters' permutation combinatorics is reused through the bridge lemma (Lemma 71).

Definition 65 (Pair partition)**.** A *pairing* of $\mathrm{Fin}\,(2n)$ is a fixed-point-free involution p of $\mathrm{Fin}\,(2n)$; the canonical *chords* of p are the pairs $(a, p\,a)$ with $a < p\,a$. There are $(2n-1)!!$ pairings.

Definition 66 (Loops)**.** For pairings p, q, the union of their chord systems decomposes into closed loops; $\ell(p, q)$ is their number, equal to $c(pq)/2$ with c the full cycle count of Definition 6 (each loop of $2m$ points contributes two m-cycles to the product pq).

Definition 67 (Crossings)**.** $\mathrm{cr}(p)$ is the number of pairs of chords $(a, p\,a)$, $(c, p\,c)$ with $a < c < p\,a < p\,c$ — the crossing number of the chord diagram.

Definition 68 (Crossing parity)**.** $\varepsilon(p) := (-1)^{\mathrm{cr}(p)}$.

Definition 69 (Permutation-like pairings)**.** For $\rho \in S_n$, q_ρ is the pairing matching $a \leftrightarrow n + \rho(a)$ for $a < n$. These are exactly the pairings with no chord inside $\{0, \dots, n-1\}$ or inside $\{n, \dots, 2n-1\}$.

Lemma 70 (Injectivity)**.** *$\rho \mapsto q_\rho$ is injective.*

Lemma 71 (Bridge to the unitary cycle count)**.** $\ell(q_\tau, q_\rho) = c(\tau^{-1}\rho)$.

Proof. Write the $2n$ points as a *top* block $\{0, \dots, n-1\}$ and a *bottom* block $\{n, \dots, 2n-1\}$. By definition (Def. 69) every chord of q_ρ joins a top point a to the bottom point $n + \rho(a)$, and involutivity forces the partner formula $q_\rho(n+b) = \rho^{-1}(b)$. Thus every vertex carries one cross-chord of q_τ and one of q_ρ, so each joint loop of the union alternates between top and bottom and visits at least one top point.

The step map on top points. Trace a loop from a top point a, traversing first the q_τ-chord and then the q_ρ-chord:

$$a \xrightarrow{q_\tau} n+\tau(a) \xrightarrow{q_\rho} q_\rho(n+\tau(a)) = \rho^{-1}(\tau(a)) = (\rho^{-1}\tau)(a).$$

One full q_τ-then-q_ρ step returns to the top block and acts as $a \mapsto (\rho^{-1}\tau)(a)$. Iterating, the top points met along the loop through a are exactly the orbit $a, (\rho^{-1}\tau)(a), (\rho^{-1}\tau)^2(a), \dots$ of a under $\rho^{-1}\tau \in S_n$; the loop closes exactly when this orbit closes (the bottom points $n+\tau(x)$ are determined by the top ones).

Loops correspond to orbits. The joint loops partition all $2n$ points, hence partition the top block. By the step map the top points of each loop form one orbit of $\rho^{-1}\tau$, and conversely every orbit is realized by exactly one loop. This bijection gives $\ell(q_\tau, q_\rho) = c(\rho^{-1}\tau)$. Finally $\rho^{-1}\tau$ and $\tau^{-1}\rho = (\rho^{-1}\tau)^{-1}$ are mutually inverse, and a permutation and its inverse have identical orbits, so $c(\rho^{-1}\tau) = c(\tau^{-1}\rho)$ and $\ell(q_\tau, q_\rho) = c(\tau^{-1}\rho)$.

Reduced form. Rather than trace loops one at a time, `loops_qPair_qPair` computes the product permutation in one shot: through the block isomorphism `pairEquiv` the underlying `swapAcrossPerm` of $q_\tau\, q_\rho$ is the block-diagonal $\mathrm{sumCongr}(\tau^{-1}\rho,\, \tau\rho^{-1})$ (the two blocks are the top step map and its bottom mirror). Its cycle count splits additively as $c(\tau^{-1}\rho) + c(\tau\rho^{-1})$, and the inverse-symmetry lemma `cycleCount_mul_inv_comm` identifies the two summands, giving $2\,c(\tau^{-1}\rho)$; since ℓ is half the joint cycle count, an `omega` halving closes the goal. This replaces the explicit loop/orbit bijection (and the $n=4$ exhaustive check PL1) by an algebraic cycle-type computation. □

Definition 72 (Determinant vector)**.** $v \in k^{\mathcal{P}_2(2n)}$ has $v_{q_\rho} = \mathrm{sgn}(\rho)$ and $v_p = 0$ for p not permutation-like.

Definition 73 (The ε vector)**.** $\varepsilon \in k^{\mathcal{P}_2(2n)}$ has entries $\varepsilon_p = (-1)^{\mathrm{cr}(p)}$.

Definition 74 (Canonical word)**.** The *canonical word* of p is the permutation sending $(0,1,2,3,\dots)$ to $(a_0, b_0, a_1, b_1, \dots)$ where (a_i, b_i) are the chords listed with $a_i < b_i$ and $a_0 < a_1 < \cdots$.

Lemma 75 (Pfaffian sign identity)**.** $\mathrm{sgn}($*canonical word of* $p) = (-1)^{\mathrm{cr}(p)}$. *This is the sign with which p enters the Pfaffian expansion.*

Proof. The argument is a non-inductive inversion count. By the Mathlib fact $\mathrm{sgn}(\sigma) = (-1)^{\mathrm{inv}(\sigma)}$ (used through `sign_eq_prod_prod_Iio`, which writes $\mathrm{sgn}(\sigma)$ as the product over ordered position pairs $s < t$ of the per-pair inversion sign), it suffices to evaluate the inversion sign of $\sigma = \mathrm{canonicalWord}\, p$. We do this in closed form, never recursing on n.

Reindexing positions by chord coordinates. List the n canonical chords as $(a_0, b_0), \dots, (a_{n-1}, b_{n-1})$ with $a_k < b_k$ and, by the canonical listing, with left endpoints sorted $a_0 < a_1 < \cdots < a_{n-1}$ (Def. 74). The canonical word has value sequence $(a_0, b_0, a_1, b_1, \dots)$, i.e. $\sigma(2k) = a_k$ and $\sigma(2k+1) = b_k$, so a position is encoded by a pair $(\text{chord index}, \text{slot}) \in \mathrm{Fin}\, n \times \mathrm{Fin}\, 2$. Grouping the product over position pairs by the pair of chords (i,j) they involve, the four cross-slot comparisons of chords $i < j$ multiply to a single chord factor $(-1)^{\mathrm{crossInd}(i,j)}$.

The per-chord-pair case split. Fix $i < j$. All four positions of chord i precede those of chord j, so the only inversions among them come from the four cross-slot pairs with values (a_i, a_j), (a_i, b_j), (b_i, a_j), (b_i, b_j). Since $a_i < a_j$ (sorted left endpoints) and $a_i < b_i$, $a_j < b_j$, the value a_i is the global minimum, so the pairs starting from a_i are never inversions. The remaining freedom is the position of b_i relative to $a_j < b_j$, giving exactly three exclusive configurations (all $2n$ endpoints distinct):

$$\begin{aligned} &\textbf{disjoint } (a_i < b_i < a_j < b_j) &&\Rightarrow \quad 0 \text{ inversions;} \\ &\textbf{crossing } (a_i < a_j < b_i < b_j) &&\Rightarrow \quad 1 \text{ inversion;} \\ &\textbf{nesting } (a_i < a_j < b_j < b_i) &&\Rightarrow \quad 2 \text{ inversions.} \end{aligned}$$

Thus the chord factor is $(-1)^{\mathrm{crossInd}(i,j)}$ with $\mathrm{crossInd}(i, j) = 1$ exactly in the crossing case $a_i < a_j < b_i < b_j$, and 0 otherwise (out-of-order pairs $j \le i$ also contribute 1, i.e. no sign).

Summation and the order-isomorphism bridge. Multiplying over all chord pairs,

$$\mathrm{sgn}(\sigma) = \prod_{i,j} (-1)^{\mathrm{crossInd}(i,j)} = (-1)^{\sum_{i,j} \mathrm{crossInd}(i,j)}.$$

Finally $\sum_{i,j} \mathrm{crossInd}(i, j) = \mathrm{cr}(p)$ (Def. 67): the left-endpoint embedding $i \mapsto a_i$ is a strictly increasing injection onto the left endpoints, hence an order isomorphism, so it carries the chord-index crossing condition $i < j \wedge a_i < a_j < b_i < b_j$ bijectively onto the endpoint crossing condition defining $\mathrm{cr}(p)$. The nesting pairs, contributing the even count 2 each, drop out of the parity, leaving $\mathrm{sgn}(\sigma) = (-1)^{\mathrm{cr}(p)} = \varepsilon(p)$. (Exhaustive machine check: PS1, all pairings through $n = 5$.)

Reduced form. `sign_canonicalWord` runs exactly this route: `sign_chord` rewrites $\mathrm{sgn}(\mathrm{canonicalWord}\, p)$ as $\prod_i \prod_j$ `chordFactor` $i\, j$, then `chordFactor_eq_pow` discharges the three-way case split (crossing $\mapsto -1$, nesting $\mapsto +1$, disjoint/out-of-order $\mapsto 1$) to give `chordFactor` $= (-1)^{\mathrm{crossInd}}$. Two applications of `Finset.prod_pow_eq_pow_sum` collapse the double product to $(-1)^{\sum_{i,j} \mathrm{crossInd}}$, and `sum_crossInd_eq_crossings` supplies the order-isomorphism bridge as a `Finset.card_bij` along `leftEndEmb`, finishing without any induction on n. □

Chapter 11

The orthogonal Gram and its below-threshold package

The orthogonal Weingarten convention follows Collins–Śniady (*Comm. Math. Phys.* **264** (2006), 773–795): the Gram data on pairings has entries $N^{\ell(p,q)}$ and the Weingarten data is its pseudo-inverse. Zinn-Justin (arXiv:0907.2719) and Matsumoto (arXiv:1001.2345) realize this Gram as a product of odd Jucys–Murphy shifts on the pairing module, with zonal spectrum $\prod_{(i,j)\in\lambda}(N+2j-1-i)$; Matsumoto (arXiv:1004.4717) records the general-parameter function and tables (our exact $n=2$ values match his Example 1 verbatim), with the convolution-inverse identity attributed to Collins–Matsumoto (arXiv:0903.5143). The signed sum *above* threshold is in print: Campbell–Yin (arXiv:1907.01009) display the quadrature value $(d-n)!/d!$ via the zonal evaluation $\sum_\sigma \operatorname{sgn}(\sigma)\,\omega^\lambda(\sigma) = ((n+1)!/2^n)\,\delta_{\lambda,1^n}$, an instance of the Marcus–Spielman–Srivastava minor-orthogonality phenomenon; the bookend zonal values are Macdonald VII.2.2(b) and VII.2.25. What we have *not* located displayed: the below-threshold vanishing as a stated identity, the absorption identities in eigenrelation form, and the Penrose-conditional route — all presumed routine; the reference request covers them. The proofs below are complete and elementary; suites BO1–BO4 and PL1–PL8 check every step exhaustively through $n=4$ (the 105×105 cell included) in exact arithmetic.

Definition 76 (Orthogonal Gram matrix)**.** $G_n^{\mathrm{O}}(N) \in k^{\mathcal{P}_2(2n)\times\mathcal{P}_2(2n)}$ has entries $G_n^{\mathrm{O}}(N)_{pq} = N^{\ell(p,q)}$, for any commutative ring k and any $N\in k$.

Definition 77 (Even-rising factorial)**.** $r_n(N) := N(N+2)\cdots(N+2n-2)$.

Definition 78 (Falling factorial)**.** $f_n(N) := N(N-1)\cdots(N-n+1)$.

Theorem 79 (All-ones absorption)**.** $G_n^{\mathrm{O}}(N)\,\mathbf{1} = r_n(N)\,\mathbf{1}$.

Proof. The p-th entry of $G_n^{\mathrm{O}}(N)\,\mathbf{1}$ is the row sum $R_n(N) := \sum_{q\in\mathcal{P}_2(2n)} N^{\ell(p,q)}$ (Def. 66, Def. 76). We show $R_n(N)$ is independent of p and equals the even-rising factorial $r_n(N) = N(N+2)\cdots(N+2n-2)$.

Constancy. S_{2n} acts on $\mathcal{P}_2(2n)$ by relabeling $g\cdot p := g\,p\,g^{-1}$, again a fixed-point-free involution. All such involutions have cycle type 2^n, so the action is transitive, and the chord union of $(g\cdot p, g\cdot q)$ is the g-image of that of (p,q), whence $\ell(g\cdot p, g\cdot q) = \ell(p,q)$. Reindexing the sum by $q = g\cdot q'$ gives $\sum_q N^{\ell(g\cdot p,q)} = \sum_{q'} N^{\ell(g\cdot p, g\cdot q')} = \sum_{q'} N^{\ell(p,q')}$, so the row sum at $g\cdot p$ equals that at p. By transitivity every row sum coincides; write the common value $R_n(N)$.

Recursion. $R_0(N) = 1$: $\mathscr{P}_2(0)$ has the single empty pairing with term $N^0 = 1$. For $n \geq 1$ fix p, distinguish a point z with p-partner $w := p(z)$, and sum over q by conditioning on $y := q(z)$. If $y = w$ then $\{z, w\}$ is a double edge, a 2-loop on its own; deleting z, w identifies q with an arbitrary pairing of the remaining $2n-2$ points and gives $\ell(p, q) = 1 + \ell(p\restriction, q\restriction)$, so this case contributes $N \cdot R_{n-1}(N)$. For each of the $2n-2$ choices $y \neq w$ (the splice), put $u := p(y)$; the four points z, w, y, u are distinct, and in the chord union z, y are interior degree-two vertices on the path $w \stackrel{p}{-} z \stackrel{q}{-} y \stackrel{p}{-} u$. Replacing the p-chords $\{z, w\}, \{y, u\}$ by the single chord $\{w, u\}$ and deleting z, y contracts these two degree-two vertices — a homeomorphism of the 1-complex — so the loop count is unchanged, $\ell(p, q) = \ell(\tilde{p}, \tilde{q})$ (in the boundary case $q(u) = w$ the 4-loop becomes a 2-loop, still one loop; $\{z, w\}$ is never a double edge here since $q(z) = y \neq w$). As q ranges over pairings with $q(z) = y$, its restriction ranges over all pairings of the $2n-2$ points, contributing $R_{n-1}(N)$ by constancy at level $n-1$. Summing the two cases,

$$R_n(N) = N\, R_{n-1}(N) + (2n-2)\, R_{n-1}(N) = (N + 2n - 2)\, R_{n-1}(N).$$

Iterating from $R_0 = 1$ yields $R_n(N) = \prod_{j=1}^{n}(N + 2j - 2) = r_n(N)$ (Def. 77). Every entry of $G_n^{\mathrm{O}}(N)\,\mathbf{1}$ thus equals $r_n(N)$. As both sides are integer-coefficient polynomials in N, the identity holds over any commutative ring. (Machine check: PL5 through $n = 5$, 945 pairings.)

Reduced form. `orthGram_mulVec_one` reduces each entry to `rowSum` via `rowSum_eq` and then evaluates it by `rowSum_eq_evenRising`. Constancy is `rowSum_const`: any two pairings are conjugate (`pairing_isConj`), and the conjugation bijection `conjPEquiv` reindexes the sum while `loops_conjP` preserves the exponent. The recursion `rowSum_succ` distinguishes the point 0 and splits $\mathscr{P}_2(2(m+1))$ along the bijection `biEquiv` into the close fiber (`closeP`, with `loops_closeP` giving the extra factor N) and the splice fiber indexed by $\mathrm{Fin}\,2m \times \mathscr{P}_2(2m)$ (`spliceP`, with `loops_spliceP` giving loop-count invariance); each fiber sums to $\texttt{rowSum}_m$, and `ring` assembles $(N + 2m)\,\texttt{rowSum}_m$. The base $r_0 = 1$ is `rowSum_zero`. ☐

Theorem 80 (Determinant-vector absorption). $G_n^{\mathrm{O}}(N)\, v = f_n(N)\, v$.

Proof. The p-th entry of $G_n^{\mathrm{O}}(N)\, v$ is

$$(G^{\mathrm{O}} v)_p = \sum_q N^{\ell(p,q)}\, v_q = \sum_{\rho \in S_n} \mathrm{sgn}(\rho)\, N^{\ell(p,\, q_\rho)},$$

the sum collapsing to the permutation-like rows $q = q_\rho$ because v vanishes elsewhere (Def. 72). We show $(G^{\mathrm{O}} v)_p = f_n(N)\, v_p$ for every row p, in two cases.

On the permutation block: $p = q_\tau$. By the bridge lemma 71, $\ell(q_\tau, q_\rho) = c(\tau^{-1}\rho)$. Substituting $\pi := \tau^{-1}\rho$ (a bijection of S_n as ρ varies, so $\rho = \tau\pi$ and $\mathrm{sgn}(\rho) = \mathrm{sgn}(\tau)\,\mathrm{sgn}(\pi)$) gives

$$(G^{\mathrm{O}} v)_{q_\tau} = \sum_\rho \mathrm{sgn}(\rho)\, N^{c(\tau^{-1}\rho)} = \mathrm{sgn}(\tau) \sum_\pi \mathrm{sgn}(\pi)\, N^{c(\pi)} = \mathrm{sgn}(\tau)\, f_n(N),$$

the last equality being the signed character of the Gram element (Lemma 12, $\sum_\pi \mathrm{sgn}(\pi)\, N^{c(\pi)} = f_n(N)$). Since $v_{q_\tau} = \mathrm{sgn}(\tau)$ this equals $f_n(N)\, v_{q_\tau}$, as required.

Off the block: p not permutation-like. Then $v_p = 0$, so it suffices to show $(G^{\mathrm{O}} v)_p = \sum_\rho \mathrm{sgn}(\rho)\, N^{\ell(p,q_\rho)} = 0$. *(Block balance.)* Writing t, s, x for the numbers of top-internal, bottom-internal, and cross chords of p, counting the n top and the n bottom points gives $2t + x = n =$

$2s + x$, hence $t = s$; so p is permutation-like iff $t = 0$, and otherwise $t = s \geq 1$ and p has a top-internal chord $\{a, b\} \subseteq \{0, \dots, n-1\}$, $a \neq b$. Fix it and consider the involution

$$\iota : S_n \to S_n, \qquad \iota(\rho) = \rho \circ (a\,b).$$

It is fixed-point-free with $\iota \circ \iota = \mathrm{id}$ ($(a\,b) \neq \mathrm{id}$), and sign-reversing: $\mathrm{sgn}(\iota(\rho)) = \mathrm{sgn}(\rho)\,\mathrm{sgn}((a\,b)) = -\,\mathrm{sgn}(\rho)$. We claim it preserves the loop count,

$$\ell(p,\, q_{\iota(\rho)}) = \ell(p,\, q_\rho). \tag{11.1}$$

Writing $\rho' := \iota(\rho)$, the pairings $q_\rho, q_{\rho'}$ differ only in the two chords at a, b: $\{a, n+\rho(a)\}, \{b, n+\rho(b)\}$ versus $\{a, n+\rho(b)\}, \{b, n+\rho(a)\}$. In the chord union with p the fixed p-chord $\{a, b\}$ makes a, b interior degree-two vertices of a path with the *same* unordered endpoints $\{n+\rho(a), n+\rho(b)\}$ in both unions, only "crossed." Since a, b are top points while $q_\rho, q_{\rho'}$ send them to bottom points, $\{a, b\}$ is never a double edge, so the path has length ≥ 3 with distinct endpoints; contracting the two degree-two vertices replaces each path by the same single edge $n + \rho(a) - n + \rho(b)$ and leaves all other chords untouched. Contraction of degree-two vertices preserves the loop count, giving (11.1). The orbits $\{\rho, \rho'\}$ of ι then cancel in pairs: $\mathrm{sgn}(\rho) N^{\ell(p, q_\rho)} + \mathrm{sgn}(\rho') N^{\ell(p, q_{\rho'})} = 0$, so $(G^{\mathrm{O}} v)_p = 0 = f_n(N)\, v_p$.

Both cases give $(G^{\mathrm{O}} v)_p = f_n(N)\, v_p$, hence $G_n^{\mathrm{O}}(N)\, v = f_n(N)\, v$. Both sides are polynomial in N, so the identity holds over any commutative ring.

Reduced form. `orthGram_mulVec_detVec` does `funext p` and a `by_cases` on whether $p = q_\tau$. The on-block entry is `mulVec_detVec_perm_case`: rewrite to the signed loop sum (`mulVec_detVec_entry`), apply the bridge `loops_qPair_qPair` (Lemma 71), reindex by `Equiv.mulLeft` τ to peel off $\mathrm{sgn}(\tau)$, and close with the Gram character `sum_sgn_pow_cycleCount` (Lemma 12). The off-block entry is `signed_loop_sum_eq_zero`: `exists_topchord` supplies $\{a, b\}$ via the block-balance count, then `Finset.sum_involution` on $\rho \mapsto \rho \cdot (a\,b)$ finishes, its loop-invariance hypothesis being `loops_preserved` (a conjugation `cycleCount_conj` realizing the degree-two contraction) and its sign flip `Equiv.Perm.sign_swap`. □

Definition 81 (Commuting Penrose partner)**.** Matrices G, W are *commuting Penrose partners* if $GWG = G$, $WGW = W$, and $WG = GW$. This is the matrix-level analogue of Definition 19; the commutation, automatic in the one-algebra setting of Chapter 4, is added as a hypothesis here and holds for the Moore–Penrose inverse of a symmetric matrix and for every polynomial-in-G partner.

Lemma 82 (Per-eigenvalue Penrose rule)**.** *Let W be a commuting Penrose partner of G over a commutative ring, and let $y^\top G = \lambda\, y^\top$ with λ a unit. Then $\lambda\,(y^\top W) = y^\top$. No invertibility of G is assumed: only the single eigenvalue must be a unit.*

Proof. Multiplying the eigen-row into $GWG = G$ gives $\lambda\,(y^\top WG) = \lambda\, y^\top$; cancel the unit λ, then $WG = GW$ turns $y^\top GW$ back into $\lambda\, y^\top W$. Division-free: the conclusion is in λ-multiplied form. □

Lemma 83 (Two-line evaluation)**.** *If W is a commuting Penrose partner of $G = G_n^{\mathrm{O}}(N)$ then $v^\top W = f_n(N)\,(v^\top W^2)$.*

Proof. $v^\top W = v^\top(WGW) = v^\top G W^2 = (Gv)^\top W^2 = f_n(N)\, v^\top W^2$, using $WGW = W$, the commutation, the symmetry of G, and Theorem 80. □

Theorem 84 (Exact vanishing of the signed sums). *If $f_n(N) = 0$ — in particular at every integer $1 \le N \le n-1$ — then $v^\mathsf{T} W = 0$ for every commuting Penrose partner W: all signed determinant sums of the orthogonal Weingarten data vanish exactly.*

Proof. Immediate from Lemma 83. (Machine check: BO4 at $n = 3$, $N = 1, 2$, with Penrose-certified exact pseudo-inverses.) □

Theorem 85 (Kernel witness). *If $f_n(N) = 0$ then $G_n^{\mathrm{O}}(N)\, v = 0$.*

Theorem 86 (Gram singularity below threshold). *If $f_n(N) = 0$ and $v \neq 0$ then $G_n^{\mathrm{O}}(N)$ is not invertible. In particular, over the* nonnegative *integers the invertibility threshold is $N \ge n$: singularity at $0 \le N < n$ is this node's machine-checked content, while invertibility at integer $N \ge n$ is the known spectral endpoint (the zonal spectrum cited in this chapter's preamble) and is* not *formalized here. The nonnegativity scoping matters: at negative integers the zonal factors $\prod_{(i,j)\in\lambda}(N + 2j - i - 1)$ can be nonzero (already $G_2^{\mathrm{O}}(-1)$ has determinant -4 and is invertible over $\mathbb{Q}$), so no unqualified integer threshold statement holds.*

Theorem 87 (Conditional even-rising rule). *If $G_n^{\mathrm{O}}(N)$ is invertible and W is a commuting Penrose partner, then $r_n(N)\, \mathbf{1}^\mathsf{T} W = \mathbf{1}^\mathsf{T}$: every row sum of the orthogonal Weingarten data is $1/r_n(N)$.*

Proof. $GWG = G$ with G invertible forces $W = G^{-1}$, and $\mathbf{1}^\mathsf{T} = \mathbf{1}^\mathsf{T} G G^{-1} = r_n(N)\, \mathbf{1}^\mathsf{T} G^{-1}$ by symmetry and Theorem 79. □

Theorem 88 (Conditional falling rule). *Under the same hypotheses, $f_n(N)\, v^\mathsf{T} W = v^\mathsf{T}$: the signed determinant sums invert the falling factorial above threshold, matching the displayed Campbell–Yin value.*

Theorem 89 (Per-eigenvalue even-rising rule). *If W is a commuting Penrose partner of $G_n^{\mathrm{O}}(N)$ and $r_n(N)$ is a unit, then $r_n(N)\, \mathbf{1}^\mathsf{T} W = \mathbf{1}^\mathsf{T}$ — even where the Gram itself is singular (e.g. $n = 2$, $N = 1$). This strengthens Theorem 87: full invertibility of the Gram is replaced by the single relevant eigenvalue.*

Proof. Lemma 82 with the eigen-row $\mathbf{1}^\mathsf{T} G = r_n(N)\, \mathbf{1}^\mathsf{T}$ (Theorem 79 plus symmetry). (Machine check: `verify_penrose_eigen.py` at the singular cell $n = 2$, $N = 1$.) □

Theorem 90 (Per-eigenvalue falling rule). *Under the same partner hypothesis, if $f_n(N)$ is a unit then $f_n(N)\, v^\mathsf{T} W = v^\mathsf{T}$ — complementary to the exact vanishing of Theorem 84 when $f_n(N) = 0$.*

Proof. Lemma 82 with the eigen-row of Theorem 80. □

Remark 91 (Integral reading). For Haar $O \in \mathrm{O}(N)$ with $N \ge n$ and A the top-left $n \times n$ corner, $E[(\det A)^2] = n!\, (v^\mathsf{T}\mathrm{Wg})_{q_{\mathrm{id}}} = n!/f_n(N)$ above threshold. Below threshold ($N < n$) no $n \times n$ corner of an $N \times N$ matrix exists; the surviving object is the $n \times n$ array $A_{ab} := O_{\iota(a)\iota(b)}$ built from index maps $\iota : [n] \to [N]$ — any such array has a repeated row, so $\bigwedge^n \mathbb{R}^N = 0$ forces $\det A = 0$ pointwise — and Theorem 84 is exactly the Weingarten shadow of that rank deficiency (the same nonexistence-of-the-corner convention as Remark 111). The *orthogonal* Haar moment formula itself remains out of Lean scope (Chapter 14): the machine-checked Haar bridge (Chapter 13) now in-repo under `Weingarten/Haar/` covers $\mathrm{U}(N)$ at the formalized degrees only, so this reading is still recorded as prose only.

Chapter 12

The symplectic Gram: crossing-sign factorization and the role swap

The symplectic series $\mathrm{Sp}(M) \subset \mathrm{U}(2M)$ is conventionally reached from the orthogonal one by a parameter twist. The one-argument relation $\mathrm{Wg}^{\mathrm{Sp}}(\sigma, M) = \pm \mathrm{Wg}^{\mathrm{O}}(\sigma, -2M)$ on the Weingarten function is in print — Redelmeier (arXiv:1412.0646, Remark 6.8) gives the sign as $(-1)^{n/2}$, Coulter–Do (arXiv:2506.04002, Remark 2.13) as a $\pm$ — and the negative-dimension duality $\mathrm{Sp}(M) \leftrightarrow \mathrm{O}(-2M)$ is a classical genre. A *two-argument*, Gram-level symplectic statement is also in print: Matsumoto (arXiv:1301.5401, §2.3.2) builds the symplectic Weingarten function through the twisted Gelfand pair $(S_{2k}, H_k, \varepsilon)$, with $T^{\mathrm{Sp}}(\sigma; z) = (-1)^k \varepsilon(\sigma)(-2z)^{\ell(\mu)}$ (ε the signature), $\mathrm{Wg}^{\mathrm{Sp}}$ its pseudo-inverse (eq. (2.8)), and $\mathrm{Wg}^{\mathrm{Sp}}(\sigma; z) = (-1)^k \varepsilon(\sigma) \mathrm{Wg}^{\mathrm{O}}(\sigma; -2z)$; his Theorem 2.4 and eq. (2.15) display the Gram itself as the matrix of J-contractions $\mathrm{G} = (T^{\mathrm{Sp}}(\sigma^{-1}\tau))$ (eq. (2.14): $T_\sigma(J, J) = T^{\mathrm{Sp}}(\sigma; n)$). Redelmeier's Definition 6.7 gives a second two-argument form $\mathrm{Gr}(\pi_+, \pi_-) = (-1)^{n/2}(-2N)^{\#(\pi_+ \vee \pi_-)}$, but with a *uniform* sign, the per-pairing crossing parity absorbed into the basis. What we provide is therefore not the object but an elementary, character/zonal-free *proof* of it: the *canonical-contraction* form $\mathrm{G}^{\mathrm{Sp}}(\pi, \sigma) = \varepsilon(\pi)\varepsilon(\sigma)(-1)^n(-2M)^{\#\mathrm{loop}(\pi,\sigma)}$ (the factorization below), with the crossing parity ε exposed entrywise (Matsumoto's quotient signature $\varepsilon(\sigma^{-1}\tau)$ factored via the homomorphism property, the canonical representative's signature identified with the crossing parity). What we have not located displayed separately is this crossing-parity-exposed presentation, the below-threshold ε-row vanishing, or the expected-Pfaffian compression formula of Remark 111. No novelty is inferred; the design note carries the ledger. Ground truth: all 256 degree-four and all 4096 degree-six Haar moments of $\mathrm{Sp}(1) = \mathrm{SU}(2)$, integrated exactly, match the Weingarten formula built on this Gram (suite A5); the crossing-sign argument is given in full at Theorem 97 below, pinned by suites PS1–PS6 and PF0–PF4.

Definition 92 (Standard form)**.** J on $\mathrm{Fin}\,(2M)$ has $J_{x,x+M} = 1$, $J_{x+M,x} = -1$, all other entries 0; $J^2 = -1$.

Definition 93 (Canonical contraction)**.** For a pairing p and an index function $i : \mathrm{Fin}\,(2n) \to \mathrm{Fin}\,(2M)$, $\Delta'_p(i) := \prod_{(a,\, p\, a),\, a < p\, a} J_{i(a),\, i(p\, a)}$ — the product over canonical chords, well defined without any word choice.

Definition 94 (Symplectic Gram, contraction form)**.** $G^{\mathrm{Sp}}_{pq} := \sum_i \Delta'_p(i)\,\Delta'_q(i)$, the sum over all index functions.

Definition 95 (Symplectic Gram, closed form)**.** $\widehat{G}^{\mathrm{Sp}}_{pq} := \varepsilon(p)\,\varepsilon(q)\,(-1)^n\,(-2M)^{\ell(p,q)}$, over any commutative ring.

Lemma 96 (Loop trace)**.** $\operatorname{tr}(J^{2m}) = (-1)^m\,2M$.

Proof. $J^2 = -1$, so $J^{2m} = (-1)^m\,\mathrm{id}$ on a $2M$-dimensional space. □

Theorem 97 (Crossing-sign factorization)**.** $G^{\mathrm{Sp}} = \widehat{G}^{\mathrm{Sp}}$.

Proof. Write $\ell := \ell(p,q)$. The argument has two halves: a gauge that removes the canonical orientation choice, and a loop-by-loop evaluation of the resulting index sum.

A gauge-invariant contraction. Call $w \in S_{2n}$ a *word for* p if each consecutive value-pair $\{w(2k), w(2k+1)\}$ is a chord of p, so the n pairs list the n chords once each, in some order and orientation. For such a w set $\Phi^w_p(i) := \operatorname{sgn}(w)\prod_{k=0}^{n-1} J_{i(w(2k)),\,i(w(2k+1))}$. Relative to the canonical word σ_p (Def. 74), an arbitrary word is $w = \sigma_p \circ g$ where g permutes the n size-two blocks and flips the orientation of a subset S of chords. A block permutation, viewed in S_{2n} via the point-doubling embedding $S_n \hookrightarrow S_{2n}$, is even (each block transposition is a product of two position-transpositions), while each orientation flip is one position-transposition; hence $\operatorname{sgn}(g) = (-1)^{|S|}$, so $\operatorname{sgn}(w) = \varepsilon(p)\,(-1)^{|S|}$ using Lemma 75. On the product side, reordering whole chords only reorders scalar factors, and reversing a chord $c \in S$ replaces $J_{i(a),i(b)}$ by $J_{i(b),i(a)} = -J_{i(a),i(b)}$ by antisymmetry of J; thus the product picks up $(-1)^{|S|}$. The two sign powers cancel, giving $\Phi^w_p(i) = \varepsilon(p)\,\Delta'_p(i)$ for every word w. In particular $\Phi_p := \Phi^w_p$ is independent of w, and $\Delta'_p = \varepsilon(p)\,\Phi_p$ since $\varepsilon(p) = \pm 1$. Consequently

$$G^{\mathrm{Sp}}_{pq} = \sum_i \Delta'_p(i)\Delta'_q(i) = \varepsilon(p)\varepsilon(q)\sum_i \Phi_p(i)\Phi_q(i),$$

and it suffices to show $\sum_i \Phi_p(i)\Phi_q(i) = (-1)^n(-2M)^\ell$.

Loop-adapted words. The chords of p and q together form ℓ disjoint joint loops (Def. 66); a loop L_t on $2m_t$ points alternates p- and q-edges, $\sum_t m_t = n$. Traverse L_t as $x^t_0 \xrightarrow{p} x^t_1 \xrightarrow{q} x^t_2 \xrightarrow{p} \cdots$, closing on a q-edge. Choose the word w_p listing, loop after loop, the p-chords forward, value sequence $(x^t_0, x^t_1, \dots, x^t_{2m_t-1})$ on the block of L_t; and w_q on the *same* block allocation listing the q-chords forward, $(x^t_1, x^t_2, \dots, x^t_{2m_t-1}, x^t_0)$. Both are genuine words since the loops partition $\mathrm{Fin}\,(2n)$.

The index sum is one trace per loop. In this gauge every edge is oriented forward along its loop, so $\Phi_p(i)\Phi_q(i) = \operatorname{sgn}(w_p)\operatorname{sgn}(w_q)\prod_{t=1}^{\ell} J_{i(x^t_0),i(x^t_1)} J_{i(x^t_1),i(x^t_2)} \cdots J_{i(x^t_{2m_t-1}),i(x^t_0)}$, the $2m_t$ forward edges of L_t. The index variables are disjoint across loops, so the sum over i factors, and for one loop the cyclic chain is a trace:

$$\sum_{i|_{L_t}} J_{i(x^t_0),i(x^t_1)} \cdots J_{i(x^t_{2m_t-1}),i(x^t_0)} = \operatorname{tr}\!\left(J^{2m_t}\right) = (-1)^{m_t}\,2M,$$

by iterating $\sum_y J_{\cdot y}J_{y\cdot} = (J^2)_{\cdot\cdot}$ and then Lemma 96. Multiplying over the ℓ loops, $\sum_i \prod_{\mathrm{fwd}} J = (-1)^{\sum_t m_t}(2M)^\ell = (-1)^n(2M)^\ell$.

The alignment sign is one even cycle per loop. Put $\tau := w_q \circ w_p^{-1}$, so $\operatorname{sgn}(\tau) = \operatorname{sgn}(w_p)\operatorname{sgn}(w_q)$. Reading off the value sequences, $\tau(x^t_s) = w_q(s) = x^t_{s+1 \bmod 2m_t}$, so on the $2m_t$ points of L_t the permutation τ is the single cycle $(x^t_0\, x^t_1\, \cdots\, x^t_{2m_t-1})$ of even length, sign -1. As the loops partition the points, τ is a disjoint product of ℓ such cycles, so $\operatorname{sgn}(w_p)\operatorname{sgn}(w_q) = \operatorname{sgn}(\tau) = (-1)^\ell$ — this is the sketch's "one shift per loop": the cyclic shift aligning w_q to w_p is the $2m_t$-cycle, and an even cycle is odd.

Assembly. Combining the two displays, $\sum_i \Phi_p(i)\Phi_q(i) = (-1)^\ell(-1)^n(2M)^\ell = (-1)^n(-2M)^\ell$, whence $G^{\mathrm{Sp}}_{pq} = \varepsilon(p)\varepsilon(q)(-1)^n(-2M)^\ell = \widehat{G}^{\mathrm{Sp}}_{pq}$ for all p, q. (Verified at five cells through the 105×105 case, A2, PS1–PS6; walk-form corollary $D \equiv \operatorname{cr}(p) + \operatorname{cr}(q) + \ell \pmod 2$ exhaustive at $n \le 4$; the argument is given in full below.)

Reduced form. In Lean, `sympGramContr_eq_closed` is the thin top of this tower: after `ext p q` it rewrites $\Delta'_p\Delta'_q = \varepsilon(p)\varepsilon(q)\,\Phi_p\Phi_q$ via `deltaContr_eq_eps_PhiCanon` (which is exactly the gauge lemma, $\operatorname{sgn}(\sigma_p)^2 = 1$ collapsing the two sign powers), pulls the constants out of the sum with `Finset.mul_sum`, and closes by `ring`. The loop content is `PhiCanon_sum`, which reduces to `deltaContr_gram_closed`. Rather than factor the index sum over loops directly (the orbit-factorization above has no off-the-shelf Mathlib form), Lean proves the latter by a *merge-induction* on n: `deltaContr_gram_after_collapse` sums out vertex 0 and passes to `gram_recursion`, which splits on whether p and q share the chord at 0 — the two-loop case (`two_loop_residual`, loop count $+1$) and the merge case (`merge_residual`, loop count invariant), the merge sign being the crossing-parity leaf `qStar_sign`. The per-loop value $\operatorname{tr}(J^{2m}) = (-1)^m\, 2M$ is `trace_Jform_pow`. ∎

Corollary 98 (Conjugation form)**.** *Entrywise, $\widehat{G}^{\mathrm{Sp}}_{pq} = \varepsilon(p)\left((-1)^n\, G^{\mathrm{O}}_n(-2M)_{pq}\right)\varepsilon(q)$: the symplectic Gram is an explicit signed conjugation of the orthogonal Gram at parameter $-2M$.*

Definition 99 (ε-twisted determinant vector)**.** $(\varepsilon \odot v)_p := \varepsilon(p)\, v_p$.

Definition 100 (Even-falling factorial)**.** $e_n(M) := (2M)(2M-2)\cdots(2M-2n+2)$.

Definition 101 (Shifted rising factorial)**.** $s_n(M) := (2M)(2M+1)\cdots(2M+n-1)$.

Theorem 102 (ε-twisted ones absorption)**.** $\widehat{G}^{\mathrm{Sp}}\,\varepsilon = e_n(M)\,\varepsilon$.

Proof. The conjugation corollary (Corollary 98) gives the matrix factorization

$$\widehat{G}^{\mathrm{Sp}} = (-1)^n\, D_\varepsilon\, G^{\mathrm{O}}_n(-2M)\, D_\varepsilon,$$

where $D_\varepsilon = \operatorname{diag}(\varepsilon(p))_p$ is an involution ($\varepsilon(p)^2 = 1$). Hence $\widehat{G}^{\mathrm{Sp}}\,\varepsilon = (-1)^n\, D_\varepsilon\, G^{\mathrm{O}}_n(-2M)\, D_\varepsilon\, \varepsilon$. The vector $D_\varepsilon\varepsilon$ has p-entry $\varepsilon(p)\,\varepsilon(p) = \varepsilon(p)^2 = 1$, so $D_\varepsilon\varepsilon = \mathbf{1}$. Theorem 79 is an identity of polynomials in the parameter, valid over any commutative ring and in particular at $-2M$, so $G^{\mathrm{O}}_n(-2M)\,\mathbf{1} = r_n(-2M)\,\mathbf{1}$. Multiplying back by D_ε scales out, $(D_\varepsilon\mathbf{1})_p = \varepsilon(p)$, giving $D_\varepsilon(r_n(-2M)\mathbf{1}) = r_n(-2M)\,\varepsilon$. Collecting,

$$\widehat{G}^{\mathrm{Sp}}\,\varepsilon = (-1)^n\, r_n(-2M)\,\varepsilon.$$

The twist arithmetic finishes: writing the rising factorial at $-2M$ in linear factors,

$$r_n(-2M) = \prod_{k=0}^{n-1}(-2M+2k) = \prod_{k=0}^{n-1}(-2)(M-k) = (-2)^n\prod_{k=0}^{n-1}(M-k),$$

so $(-1)^n r_n(-2M) = 2^n\prod_{k=0}^{n-1}(M-k) = \prod_{k=0}^{n-1}(2M-2k) = e_n(M)$. Thus $\widehat{G}^{\mathrm{Sp}}\,\varepsilon = e_n(M)\,\varepsilon$.

Reduced form. `sympGram_mulVec_epsVec` rewrites with `sympGramClosed_factor` (the factorization above), peels the two D_ε layers by `diag_mulVec_epsVec` ($D_\varepsilon \varepsilon = \mathbf{1}$) and `diag_mulVec_one`, applies `orthGram_mulVec_one` at parameter $-2M$, and closes by `funext`, `simp`, and a `rw` with the scalar identity $(-1)^n\, r_n(-2M) = e_n(M)$, supplied by `evenRising_neg` together with the $(-1)^{2n} = 1$ collapse. $\square$

Theorem 103 (ε-twisted determinant absorption)**.** $\widehat{G}^{\mathrm{Sp}}\,(\varepsilon \odot v) = s_n(M)\,(\varepsilon \odot v)$.

Proof. The argument is identical to Theorem 102 with the determinant vector in place of the all-ones vector. By Corollary 98,

$$\widehat{G}^{\mathrm{Sp}}\,(\varepsilon \odot v) = (-1)^n\, D_\varepsilon\, G_n^{\mathrm{O}}(-2M)\, D_\varepsilon\,(\varepsilon \odot v).$$

The vector $D_\varepsilon(\varepsilon \odot v)$ has p-entry $\varepsilon(p) \cdot \varepsilon(p) v_p = v_p$, so $D_\varepsilon(\varepsilon \odot v) = v$. Theorem 80, polynomial in the parameter and so valid at $-2M$, gives $G_n^{\mathrm{O}}(-2M)\, v = f_n(-2M)\, v$. Pushing D_ε back, $(D_\varepsilon v)_p = \varepsilon(p) v_p = (\varepsilon \odot v)_p$, so $D_\varepsilon(f_n(-2M) v) = f_n(-2M)\,(\varepsilon \odot v)$. Hence

$$\widehat{G}^{\mathrm{Sp}}\,(\varepsilon \odot v) = (-1)^n\, f_n(-2M)\,(\varepsilon \odot v).$$

The twist arithmetic finishes: with the falling factorial at $-2M$ in linear factors,

$$f_n(-2M) = \prod_{k=0}^{n-1}(-2M - k) = (-1)^n \prod_{k=0}^{n-1}(2M + k),$$

so $(-1)^n f_n(-2M) = \prod_{k=0}^{n-1}(2M + k) = s_n(M)$, giving $\widehat{G}^{\mathrm{Sp}}\,(\varepsilon \odot v) = s_n(M)\,(\varepsilon \odot v)$. The two eigenvalues swap qualitative roles relative to the orthogonal series: the substitution $N \mapsto -2M$ sends the rising factorial r_n to the threshold-carrying e_n and the falling factorial f_n to the never-vanishing s_n, so the twisted *ones* eigenvalue $e_n(M)$ now carries the threshold while $s_n(M) > 0$ for every integer $M \geq 1$.

Reduced form. `sympGram_mulVec_epsDetVec` mirrors the ones case: rewrite by `sympGramClosed_factor`, collapse the D_ε layers via `diag_mulVec_epsDetVec` ($D_\varepsilon(\varepsilon \odot v) = v$) and `diag_mulVec_detVec`, apply `orthGram_mulVec_detVec` at $-2M$, and finish by `funext`, `simp`, and a `rw` with $(-1)^n f_n(-2M) = s_n(M)$, supplied by `fallingFact_neg` with the $(-1)^{2n} = 1$ collapse. $\square$

Theorem 104 (Kernel witness below threshold)**.** *For integers $M < n$, $\widehat{G}^{\mathrm{Sp}}\,\varepsilon = 0$: the ε vector is an explicit kernel vector.*

Theorem 105 (Gram singularity below threshold)**.** *For natural numbers $M < n$ the symplectic Gram is not invertible (over any nontrivial ring). In particular, over the* nonnegative *integers the invertibility threshold is $M \geq n$: singularity at $0 \leq M < n$ is this node's machine-checked content, while invertibility at integer $M \geq n$ is the known spectral endpoint (via the twisted spectrum cited in this chapter's preamble) and is* not *formalized here. As in Theorem 86, the nonnegativity scoping is essential — the closed form at negative M reduces to the orthogonal Gram at $-2M > 0$, which is generically invertible.*

Theorem 106 (Exact vanishing below threshold)**.** *For integers $M < n$, $\varepsilon^{\mathsf{T}} W = 0$ for every commuting Penrose partner W of $\widehat{G}^{\mathrm{Sp}}$: all ε-row sums of the symplectic Weingarten data vanish exactly. (Machine check: A6/PS6 at five cells with Penrose-certified pseudo-inverses; rows exactly 0 below threshold.)*

Theorem 107 (Conditional even-falling rule)**.** *If $\widehat{G}^{\mathrm{Sp}}$ is invertible and W is a commuting Penrose partner, $e_n(M)\,\varepsilon^{\mathsf{T}}W = \varepsilon^{\mathsf{T}}$.*

Theorem 108 (Conditional shifted-rising rule)**.** *Under the same hypotheses, $s_n(M)\,(\varepsilon \odot v)^{\mathsf{T}}W = (\varepsilon \odot v)^{\mathsf{T}}$ — the never-vanishing side of the role swap.*

Theorem 109 (Per-eigenvalue even-falling rule)**.** *If W is a commuting Penrose partner of the symplectic Gram and the even-falling factorial $e_n(M)$ is a unit, then $e_n(M)\,\varepsilon^{\mathsf{T}}W = \varepsilon^{\mathsf{T}}$ — the per-eigenvalue strengthening of Theorem 107, with no invertibility of the Gram assumed.*

Proof. Lemma 82 with Theorem 102. □

Theorem 110 (Per-eigenvalue shifted-rising rule)**.** *If the shifted rising factorial $s_n(M) = 2M(2M+1)\cdots(2M+n-1)$ is a unit — for $M \geq 1$ over $\mathbb{Q}$ this holds at* every *cell, including the whole singular band $M < n$ — then $s_n(M)\,(\varepsilon \odot v)^{\mathsf{T}}W = (\varepsilon \odot v)^{\mathsf{T}}$: the per-eigenvalue strengthening of Theorem 108.*

Proof. Lemma 82 with Theorem 103. □

Remark 111 (Integral reading: expected Pfaffians of skew compressions). For $M \geq n$, take J-closed column and row sets C ($|C| = 2n$) and R ($|R| = 2r$), the corner $A = S[R,C]$ of Haar $S \in \mathrm{Sp}(M)$, and the skew compression $B = A^{\mathsf{T}}\tilde{J}A$. Then $E[\mathrm{Pf}(B)] = \varepsilon(n_0)\prod_{j<n}(2r - 2j)\big/\prod_{j<n}(2M-2j)$, with n_0 the partner pairing of C in slot positions (the unique column-side survivor of the Weingarten contraction). The dictionary is exact — $\det^2$ is to the sign vector as Pfaffian-of-compression is to the ε vector — and the kernel statement plays two roles: the group-side denominator above threshold, and the numerator zero for $r < n$, matching the pointwise vanishing of the Pfaffian of a rank-deficient skew matrix. For $n > M$ no J-closed column set exists — the corner does not exist — so Theorem 106 is the pseudo-inverse consistency statement, exactly parallel to $E[(\det)^2]$. The displayed first moment is a project-side derivation (recorded in the design notes), conditional on the standard $\mathrm{Sp}(M)$ Haar–Weingarten moment formula — itself outside the formalized scope — exact-checked at the cells below and *not* formalized. Verified at ten cells with brute-forced Grams and a direct SU(2) anchor (PF0–PF4); the first-moment formula was not located displayed (the Pfaffian point-process structure of symplectic truncations is well studied). $\mathrm{Sp}(M)$ Haar measure is out of Lean scope (the in-repo Haar bridge of Chapter 13 is $\mathrm{U}(N)$-only, at the formalized degrees); this remark stays prose only.

Remark 112 (Conjecture: integral reading of the ε-determinant functional). The never-vanishing functional $(\varepsilon \odot v)^{\mathsf{T}}\mathrm{Wg}^{\mathrm{Sp}} = 1/s_n(M)$ should admit a direct integral reading mirroring the orthogonal sphere moment (single-entry moments are the natural candidate). This is recorded as a *conjecture*: underived, unverified, and not scheduled for formalization.

Chapter 13

The $\mathrm{U}(N)$ Haar-integration bridge

This chapter states, as tracked blueprint nodes, the `Weingarten/Haar/` layer. *Scope honesty (the claim ceiling, verbatim from the code):* these modules provide the $\mathrm{U}(N)$ Haar-integration bridge *at the formalized degrees only* — the exact $n = 1$ entry integral, the $n = 2$ fourth-moment system, and trace-word moments through total degree 5. The general-n integral $\leftrightarrow$ Gram moment identity (the Collins–Śniady master formula) is *not* formalized in this repository, and no declaration claims or names it. Everything stated below is fully proved in Lean, *sorry*-free and axiom-clean. Throughout, $G = \mathrm{U}(N)$ is the unitary group of $N \times N$ complex matrices and E denotes an expectation on the continuous observables $C(G, \mathbb{C})$.

13.1 The abstract Haar expectation

Definition 113 (Haar expectation interface)**.** A *Haar expectation* on $\mathrm{U}(N)$ is a $\mathbb{C}$-linear functional $E\colon C(\mathrm{U}(N), \mathbb{C}) \to \mathbb{C}$ with $E[1] = 1$ that is invariant under both translation actions on the argument: $E[f(g\,\cdot)] = E[f] = E[f(\cdot\, g)]$ for every unitary g (cf. Banica–Collins, Def. 4.1, whose Haar functional is additionally *positive*; positivity is deliberately omitted from this interface, so every Haar state in their sense restricts to a Haar expectation, and integration against the constructed Haar probability measure of Section 13.4 provides a positive instance — the moment theorems below consume only linearity, normalization, and bi-invariance). The domain restriction to *continuous* observables is load-bearing: for $N \geq 1$ the same interface on all functions $\mathrm{U}(N) \to \mathbb{C}$ is uninhabited (a coboundary/orbit argument, recorded as a documented argument with the definition, *not* formalized); at $N = 0$ the group is a single point and evaluation inhabits even the all-functions interface (`haarExpectationZero`).

Lemma 114 (Charge kill)**.** *If translation of the argument by some unitary g rescales an observable, $f(g\,\cdot) = w\,f$ with $w \neq 1$, then $E[f] = 0$; likewise on the right. In particular scalar phases zI ($|z| = 1$) kill every observable of nonzero total charge (Collins–Śniady, eq. (12)).*

Proof. Invariance gives $E[f] = E[f(g\,\cdot)] = w\,E[f]$, so $(w - 1)E[f] = 0$. □

13.2 The entry integrals at $n = 1$

The observables are matrix-entry evaluations $U \mapsto U_{ij}$, the trace $U \mapsto \operatorname{Tr} U$, and the pairings $U \mapsto \operatorname{Tr}(AU)$, $U \mapsto \operatorname{Tr}(BU^*)$; the structured unitaries doing the work are diagonal sign and phase matrices, permutation matrices, and one $\pi/4$ plane rotation. Everything in this section and the next is proved *conditionally on the interface alone* — no measure theory enters until Section 13.4.

Theorem 115 (The $n = 1$ entry table)**.** *For every Haar expectation E and all indices,*

$$E[U_{ij}\,\overline{U_{kl}}] = \frac{\delta_{ik}\,\delta_{jl}}{N}.$$

Proof. Row and column sign unitaries rescale the off-diagonal cases by -1, so Lemma 114 kills them. For the diagonal case, permutation bi-invariance equalizes $E\,|U_{ij}|^2$ over all (i, j), and the row-unitarity identity $\sum_b |U_{ib}|^2 = 1$, read as an identity of observables and averaged, pins the common value to $1/N$. □

Theorem 116 (Trace moments at $n = 1$)**.** $E[\operatorname{Tr} U] = 0$ *(a scalar-phase charge kill), and for* $N \geq 1$,

$$E[|\operatorname{Tr} U|^2] = 1.$$

More generally, without any hypothesis on N, $E[\operatorname{Tr}(AU)\,\operatorname{Tr}(BU^*)] = \operatorname{Tr}(AB)/N$.

Proof. Expand the traces into entries and apply Theorem 115: the double sum collapses to $\sum_i 1/N$, respectively to $\operatorname{Tr}(AB)/N$. □

13.3 The $n = 2$ fourth-moment system

Theorem 117 (The fourth-moment table)**.** *For every Haar expectation, all $i \neq j$, and with the four index patterns stated separately in Lean:*

$$E[|U_{ii}|^2|U_{ij}|^2] = \frac{1}{N(N+1)}, \quad E[|U_{ij}|^4] = \frac{2}{N(N+1)}, \quad E[|U_{ii}|^2|U_{jj}|^2] = \frac{1}{N^2-1},$$

$$E[U_{ii}U_{jj}\,\overline{U_{ij}U_{ji}}] = -\frac{1}{N(N^2-1)}.$$

Proof. Unitarity and orthogonality contractions plus permutation equalization yield an *underdetermined* linear system in the patterns; the closing equation $A = 2B$ comes from right-invariance under the $\pi/4$ plane rotation, whose cross terms are killed by a diagonal phase. The system is then solved exactly (`linear_combination`) — character-free, and with no general moment formula anywhere. □

Theorem 118 (Fourth absolute trace moment)**.** *For $N \geq 2$: $E[|\operatorname{Tr} U|^4] = 2$, exactly at finite N. (At $N = 1$ the value is 1.)*

Proof. A universal phase-covariance lemma classifies the diagonal fourth moments $E[U_{aa}U_{bb}\overline{U_{cc}U_{dd}}]$: mismatched index pair-multisets die, and the surviving counts assemble to $2/(N+1)+2N/(N+1) = 2$ via Theorem 117. □

13.4 The Mathlib construction: unconditional for every N

Theorem 119 (Compactness)**.** $\mathrm{U}(N)$ *is compact: it is closed inside the compact set of matrices all of whose entries lie in the closed unit disc.*

Definition 120 (The Haar probability measure)**.** `haarUnitary` N is Mathlib's Haar measure on $\mathrm{U}(N)$ (under an explicitly declared Borel σ-algebra), a probability measure, right-invariant as well as left-invariant (unimodularity via uniqueness of Haar probability measures).

Theorem 121 (Inhabitation)**.** *Integration against* `haarUnitary` N *inhabits the interface of Definition 113 for* every N. *Consequently every conditional theorem above holds unconditionally of the constructed expectation.*

Proof. Linearity and normalization are integral calculus; the two invariances are Mathlib's `integral_mul_left_eq_self/right`. □

Theorem 122 (Star-reality and conjugation invariance)**.** *The constructed expectation satisfies* $E[\overline{f}] = \overline{E[f]}$ *— a genuine property of the witness, not derivable from the bare interface. Separately, for* any *Haar expectation, invariance under conjugation* $f \mapsto f(gUg^{-1})$ *is free, being the composite of the two translation invariances; matrix entries transform under diagonal-phase conjugation with computable covariance, and the trace is a class function.*

Remark 123 (Second-countability shim). Two instances (second countability of the matrix space and of its unitary subtype) close a gap at the Mathlib pin; they unlock the Borel structure on $\mathrm{U}(N) \times \mathrm{U}(N)$ and on finite powers by instance search, which the pair and product tiers below rely on. Kept as a separate module for pin-advance visibility.

13.5 Trace words and the one-matrix model

Theorem 124 (Trace words through degree five)**.** *Write* $T = \operatorname{Tr} U$. *For* $N \geq 2$ *and* $a + b \leq 5$,

$$E[T^a \overline{T}^b] = \begin{cases} 1 & a = b \in \{0, 1\}, \\ 2 & a = b = 2, \\ 0 & a \neq b. \end{cases}$$

The unbalanced cases die by the charge kill at the primitive eighth root of unity $\zeta = (1+i)/\sqrt{2}$ *(the naive* $z = i$ *fails at charge four); the balanced values are Theorems 116 and 118.*

Theorem 125 (One-matrix-model coefficients)**.** *Let* $\langle \operatorname{Tr} U \rangle_\beta$ *be the tilted expectation with weight* $e^{\beta \operatorname{Re} \operatorname{Tr} U}$, *expanded as a truncated exponential-generating-function quotient with coefficients* $w_0, \dots, w_4$. *For* $N \geq 2$*:*

$$w_1 = \tfrac{1}{2}, \qquad w_2 = w_3 = w_4 = 0,$$

equivalently the truncated quotient is $\beta/2 \pmod{\beta^5}$ *— exact finite-*N *mock-Gaussianity (at* $N = 1$, $w_3 = -1/16$*). The companion expansion gives truncated quotient* $1 + \beta^2/4 \pmod{\beta^4}$*. (The analytic readings* $\langle \operatorname{Tr} U \rangle_\beta = \beta/2 + O(\beta^5)$ *and* $\langle |\operatorname{Tr} U|^2 \rangle_\beta = 1 + \beta^2/4 + O(\beta^4)$ *are prose glosses of these formalized coefficient identities: the machine-checked content is the finite truncated quotient, and the Taylor-remainder bridge to the actual integral quotient is* not *formalized here.)*

Proof. Binomial expansion of the real-part powers into trace words, then the degree-≤ 5 table; the moment lists $m_k = E[T \cdot R^k]$ and $z_k = E[R^k]$ ($R = \operatorname{Re} \operatorname{Tr} U$) through $k \leq 4$ are $m = (0, \frac{1}{2}, 0, \frac{3}{4}, 0)$ and $z = (1, 0, \frac{1}{2}, 0, \frac{3}{4})$. Finite, analysis-free moment algebra throughout. □

13.6 The one-link kernel scalars

Definition 126 (Kernel scalars)**.** For $\beta \in \mathbb{R}$ set $Z(\beta) = \int e^{\beta \operatorname{Re}\operatorname{Tr} U}\, dU$ and $\lambda_F(\beta) = N^{-1} \int \operatorname{Re}\operatorname{Tr} U \cdot e^{\beta \operatorname{Re}\operatorname{Tr} U}\, dU$. Then $Z(\beta) > 0$ and $Z(\beta) \geq 1$ for all N, β, with the central symmetrizations Z as a cosh-average and λ_F as a sinh-average; $\lambda_F(\beta) > 0$ for $N \geq 1$, $\beta > 0$ (open positivity of Haar measure).

Theorem 127 (Strict kernel inequality)**.** *For $N \geq 1$ and* every *real β: $\lambda_F(\beta) < Z(\beta)$.*

Proof. The pointwise bound $\cosh x - t \sinh x \geq e^{-|x|} \geq e^{-|\beta| N}$ for $|t| \leq 1$, integrated — no series expansion and no differentiation under the integral. (Scope honesty, verbatim from the module: "the eigen-equation and gap layers of the development in which this layer originated are NOT ported. No completeness of characters is claimed.") ☐

13.7 Pair and product Haar

Definition 128 (Pair expectation and its witness)**.** A *pair Haar expectation* over a designated E is a normalized linear functional on $C(\mathrm{U}(N)^2, \mathbb{C})$ with per-slot translation invariance, swap covariance, and both marginals equal to E (the marginal is an explicit parameter — a recorded design decision). The product measure `haarUnitary`⊗`haarUnitary` furnishes the honest witness for every N; on the witness, product observables factorize, $E_2[(f \circ \pi_1)(g \circ \pi_2)] = E[f]\, E[g]$, and the multiplication pushforward $f \mapsto E_2[f(UV)]$ is again a Haar expectation, discharged from the structure fields alone (no Fubini). (The declarations: `PairHaarExpectation`, `pairHaarExpectation`, `pairE_prod_factorizes`, `pairE_polynomial_factorizes`, `PairHaarExpectation.compMul`, `pairHaarExpectation`

Definition 129 (Product Haar and tilted measures)**.** `productHaar` $N\ k$ is the k-fold product of `haarUnitary` (probability, open-positive, two-sided invariant, Haar), with evaluation marginals $\int f(A_i) = E[f]$, the $k = 0$ Dirac and $k = 1$ identifications, and the *triangular substitution* law: if a continuous g is blind to slot i, then $A \mapsto F(A[i \mapsto g(A) \cdot A_i])$ integrates to $\int F$. The honest tilted measures (a pair Gibbs weight $e^{\beta \operatorname{Re}\operatorname{Tr}(U^{-1}V)}$ and its chain version on $L+1$ slots) are defined here with their probability instances; β is in Chatterjee units. Verbatim from the module's scope note: "no invariant-span scope claims and no gap claims are made in this file"; and from the chain measure's own docstring: "no lattice-gauge-theory claim is made in this repository."

13.8 The Weingarten dictionary at $n = 1, 2$

The two nodes below are pure S_n algebra — they import only the sum rules of Chapter 6, no Haar module — yet they close the loop: the values of this repository's Wg match the Haar moments computed independently above, exactly as the Weingarten calculus predicts at the formalized degrees.

Theorem 130 (Wg at $n = 1$)**.** *For real $N > 0$: $\mathrm{Wg}(\sigma, N) = N^{-1}$ for the unique $\sigma \in S_1$ — matching the entry integral of Theorem 115.*

Theorem 131 (Wg at $n = 2$)**.** *For real $N > 1$:*

$$\mathrm{Wg}(e, N) = \frac{1}{N^2 - 1}, \qquad \mathrm{Wg}((0\,1), N) = -\frac{1}{N(N^2 - 1)}$$

— matching the fourth-moment table of Theorem 117 (the transposition is written in the standing 0*-indexed convention; textbook sources print it as* $(1\,2)$*).*

Proof. Solve the 2×2 linear system formed by the rising and falling sum rules over S_2 — no characters and no matrix inversion (the identity value is `wg_two_one`; the transposition value is the sibling `wg_two_swap`). Committed cross-checks: the Collins–Śniady identity $2\,\mathrm{Wg}(e) + 2\,\mathrm{Wg}((0\,1)) = 2/(N(N+1))$, sign consistency with the parity theorem, and a numeric spot check at $N = 3$. □

Remark 132 (Bridge summary). The bridge-summary module contains no new declarations: it records the vocabulary map from the bridge layer onto this repository's own objects (the unitary Gram `Weingarten.gram`, the Weingarten function `Weingarten.wg`, the orthogonal Gram `Weingarten.orthGram`, the symplectic Gram `Weingarten.sympGramClosed` on the form `Weingarten.Jform`), restates the claim ceiling, and keeps one live example consuming a Weingarten-calculus theorem, so the dependency is load-bearing rather than merely declared. The moment theorems of this chapter carry *no code dependency* on Wg; only Section 13.8 genuinely consumes the sum rules — the correspondence between the two is documentation-level, by design.

Chapter 14

Boundary: deliberately out of scope

Recorded to prevent scope creep. *In scope*, and fully proved in Lean (`leanok`, *sorry*-free, axiom-clean): the unitary spine (the Jucys factorization, the two sum rules, the parity sign theorem, and the closed-form cancellation ratio), the Penrose-conditional below-threshold package with the certified $(N, n) = (2, 3)$ cell, the pair-partition combinatorics, the full orthogonal and symplectic Gram packages of Chapters 10–12, and the $\mathrm{U}(N)$ *Haar-integration bridge at the formalized degrees* (Chapter 13: the Haar expectation inhabited for every N, the $n = 1$ and $n = 2$ entry tables, trace words through degree five with the one-matrix-model coefficients, the one-link kernel scalars, and the Weingarten values at $n = 1, 2$) — that last claim exactly as stated, never "full Haar-integration formalization".

Out of scope, by deliberate choice:

- the general-n Collins–Śniady moment formula — the expansion of $\int U_{i_1 j_1} \cdots U_{i_n j_n} \overline{U_{k_1 l_1}} \cdots \overline{U_{k_n l_n}} \, dU$ as a double sum of Kronecker-delta products weighted by Wg — together with its two supporting pillars at general n: the invariant-tensor spanning argument (Schur–Weyl surjectivity) and the projection argument. None of these is formalized, and the repository deliberately contains no declaration of that name or shape;

- the restricted character-sum definition of Wg for general $N < n$ and its identification with the Penrose partners of Chapter 4: the orthogonality-route proofs need partition-indexed S_n character theory (Murnaghan–Nakayama, hook lengths) believed absent from Mathlib and worth a separate, coordinated contribution;

- the *general* existence of a Penrose partner for arbitrary integer $1 \le N < n$: a formalizable route exists — the Gram element is symmetric in the regular representation, so its minimal polynomial is squarefree and the group inverse is a polynomial in $\mathcal{G}$ — but it is real work off the critical path, and the $(2, 3)$ cell stands as the certified instance;

- certified cells beyond $(2, 3)$: the S_4 and S_5 anchors $3/62$ and $15/958$ would need 24×24 and 120×120 exact certificates, `native_decide` territory with its trust caveat;

- the $\mathrm{O}(N)$ and $\mathrm{Sp}(M)$ Haar bridges: the integral readings of the twisted functionals (Remarks 91 and 111, and the conjecture in Remark 112) remain prose only — the ported bridge is unitary-only;

- the Baik–Rains basis and the circular-ensemble (COE/CSE) Weingarten variants: compact-symmetric-space material outside this package;
- Moore–Penrose existence theory for the orthogonal and symplectic Grams, and Brauer-algebra-internal reformulations: the proofs here live entirely in pairing language;
- lattice and analytic applications (transfer operators, spectral gaps, Wilson loops on a lattice): deliberately not part of this repository, and no such result is claimed here.

Chapter 15

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